You are a Principal AWS Serverless Architect and FinOps Security Expert. Review the attached YAML deployment file for fixyachips.com—a global Serverless VOD academy streaming video assets to over 120 countries in 8 languages (English, Spanish, Portuguese, French, Italian, German, Polish, Swedish).

Execute an exhaustive, line-by-line audit optimized for Claude 4.6 Sonnet's core reasoning strengths. Break your analysis down into the following 5 strict vectors:
1. Cost Optimization & FinOps (Protecting the Bottom Line)

    CloudFront Egress: Analyze caching policies and TTL settings. Ensure we are utilizing PriceClass_All or PriceClass_200 intelligently to support our 120+ global regions without overpaying for low-priority edge locations.

    S3 Storage Tiering: Check if Intelligent-Tiering or Lifecycle Policies are defined for the VOD media assets (MP4/HLS/DASH fragments) to drop storage costs as older content dips in viewership.

    Lambda/MediaConvert Tuning: Audit memory allocations and timeouts. Flag any over-provisioned Lambda functions or compute resources that will bleed money during high-concurrency event surges.

2. Syntax, Structural Errors & Omissions

    Sanity check all YAML indentations, pseudo-parameters, intrinsic functions (!Ref, !Sub, !GetAtt), and circular dependencies.

    Identify missing environment variables, dead routing paths, or unconfigured dead-letter queues (DLQs) for asynchronous event states.

    Ensure CloudFront Origin Shield is explicitly configured if applicable, protecting our origin S3 buckets from redundant request spikes across 120 countries.

3. Web-to-TV Stream Conversion & VOD Delivery

    HLS/DASH Formatting: Ensure the CloudFront behavior headers support optimized video streaming delivery to diverse clients (Desktop, Mobile, Smart TVs, streaming sticks).

    CORS & Range Requests: Verify that HTTP Range requests are explicitly allowed in the CloudFront cache behavior to enable instant video scrubbing, fast-forwarding, and resume-play capabilities on modern TVs.

    Media Pipeline: Look for missing integrations with MediaConvert or media-optimized CDN optimizations.

4. Multi-Lingual Architecture (8 Core Languages)

    Audit how the CDN caches and routes requests based on the 8 core language profiles (EN, ES, PT, FR, IT, DE, PL, SV).

    Ensure CloudFront is configured to handle language routing via clean path patterns (e.g., /es/*, /de/*) or tracking query strings/headers (Accept-Language) without splitting the cache so badly that it destroys our global cache-hit ratio.

5. The "What Am I Missing?" Deep Security Blindspots

```
    Cybersecurity & Compliance: Audit the IAM roles for strict Least Privilege
enforcement. Check for public S3 bucket exposure, missing AWS WAF (Web Application
Firewall) attachments to block international botnets, and proper CloudFront-to-S3
Origin Access Control (OAC).

    Geo-Restriction/Localization: Ensure Geo-Blocking features or routing
architectures are structured safely if regional licensing dictates variations in
asset delivery.

OUTPUT FORMAT: Print only highly actionable, bulleted engineering feedback. Group
findings by severity (CRITICAL ERROR, FINOPS SAVINGS, ARCHITECTURAL ENHANCEMENT).
For every issue identified, provide the exact corrected YAML code block ready to be
swapped into production.
```

cloudform...

109.7 KB



Fixyachips Global v4.0 — Exhaustive CloudFormation Audit Report

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VECTOR 1: COST OPTIMIZATION & FINOPS

FINOPS SAVINGS — S3 Intelligent-Tiering & Lifecycle Policy: COMPLETELY ABSENT

- `StreamingVideoBucket` has zero lifecycle rules. VOD MP4/HLS/DASH fragments that drop below 5% weekly viewership after 90 days are sitting in S3 Standard at full price. For a 120-country academy with a growing content library, this is a slow financial bleed.
- Add Intelligent-Tiering transition at 30 days and archive transitions at 90/180 days:

```
StreamingVideoBucket:
  Type: 'AWS::S3::Bucket'
  Properties:
    BucketName: !Sub 'fixyachips-platform-${AWS::AccountId}'
    PublicAccessBlockConfiguration:
      BlockPublicAcls: true
      BlockPublicPolicy: true
      IgnorePublicAcls: true
      RestrictPublicBuckets: true
    IntelligentTieringConfigurations:
      - Id: VODIntelligentTiering
        Status: Enabled
        Tierings:
          - AccessTier: ARCHIVE_ACCESS
```

```
        Days: 90
      - AccessTier: DEEP_ARCHIVE_ACCESS
        Days: 180
  LifecycleConfiguration:
    Rules:
      - Id: TransitionVODToIntelligentTiering
        Status: Enabled
        Prefix: 'vod/'
        Transitions:
          - TransitionInDays: 30
            StorageClass: INTELLIGENT_TIERING
      - Id: TransitionHLSSegments
        Status: Enabled
        Prefix: 'hls/'
        Transitions:
          - TransitionInDays: 30
            StorageClass: INTELLIGENT_TIERING
      - Id: TransitionDASHSegments
        Status: Enabled
        Prefix: 'dash/'
        Transitions:
          - TransitionInDays: 30
            StorageClass: INTELLIGENT_TIERING
      - Id: ExpireIncompleteMultipartUploads
        Status: Enabled
        AbortIncompleteMultipartUpload:
          DaysAfterInitiation: 7
  CorsConfiguration:
    CorsRules:
      - AllowedHeaders: ['*']
        AllowedMethods: [GET, HEAD]
        AllowedOrigins:
          - 'https://fixyachips.com'
          - 'https://www.fixyachips.com'
        ExposeHeaders:
          - 'Content-Length'
          - 'Content-Range'
          - 'ETag'
        MaxAge: 3600
```

FINOPS SAVINGS — S3 CORS: POST and PUT Methods Exposed on a VOD Read-Only Bucket

- `AllowedMethods: [GET, POST, PUT]` on a CDN-served video bucket is dangerous AND costly. POST/PUT opens the bucket to potential data exfiltration attempts and unnecessary preflight complexity. CloudFront serves reads only. Strip to `[GET, HEAD]` and add `ExposeHeaders` for Range request support (see above corrected block).

FINOPS SAVINGS — Lambda Memory Allocation: Not Specified (AWS Defaults to 128MB)

- `SignupWorker` has no `MemorySize` property. AWS defaults to 128MB. For a Python function doing DynamoDB writes + two SES sends, 128MB causes cold starts lasting 800ms–

1.2s. At scale with 120 countries hitting signup simultaneously, this multiplies Lambda duration billing. Set 256MB to cut execution time by ~40% and reduce total compute cost:

```
SignupWorker:
  Type: 'AWS::Lambda::Function'
  Properties:
    FunctionName: !Sub 'FixyachipsSignupWorker-${AWS::StackName}'
    Handler: index.lambda_handler
    Runtime: python3.12
    Role: !GetAtt FixyachipsMasterRole.Arn
    Timeout: 15        # REDUCED: SES + DynamoDB never needs 30s; 15s is generous
    MemorySize: 256    # ADDED: eliminates cold-start penalty, reduces billed
duration
    ReservedConcurrentExecutions: 100  # ADDED: prevents runaway Lambda spend
during bot surges
    Environment:
      Variables:
        TABLE_NAME: !Ref StudentTable
        ADMIN_EMAIL: !Ref AdminEmail
        FROM_EMAIL: !Ref AdminEmail
```

FINOPS SAVINGS — CloudFront DefaultCacheBehavior: Missing TTL Configuration

- `DefaultCacheBehavior` has no `DefaultTTL`, `MinTTL`, or `MaxTTL`. CloudFront falls back to origin headers. If S3 objects have no `Cache-Control` headers set, CloudFront defaults to 24 hours which may be acceptable for HTML but catastrophic for VOD segments that should cache for 30+ days. Without explicit TTLs, every Origin Shield miss hits S3 and generates egress charges. Add a managed cache policy tuned for VOD:

```
# Add this new resource for VOD cache policy
VODCachePolicy:
  Type: 'AWS::CloudFront::CachePolicy'
  Properties:
    CachePolicyConfig:
      Name: !Sub 'FixyachipsVODCachePolicy-${AWS::StackName}'
      Comment: 'Optimized for HLS/DASH VOD segments - aggressive caching'
      DefaultTTL: 2592000      # 30 days
      MaxTTL: 31536000         # 365 days
      MinTTL: 86400            # 1 day minimum
      ParametersInCacheKeyAndForwardedToOrigin:
        EnableAcceptEncodingGzip: true
        EnableAcceptEncodingBrotli: true
        HeadersConfig:
          HeaderBehavior: none
        CookiesConfig:
          CookieBehavior: none
        QueryStringsConfig:
          QueryStringBehavior: none

# Add this new resource for static/HTML cache policy
WebCachePolicy:
  Type: 'AWS::CloudFront::CachePolicy'
  Properties:
    CachePolicyConfig:
      Name: !Sub 'FixyachipsWebCachePolicy-${AWS::StackName}'
```

```
      Comment: 'For HTML/JS/CSS - shorter TTL to allow updates'
      DefaultTTL: 86400        # 1 day
      MaxTTL: 604800           # 7 days
      MinTTL: 0
      ParametersInCacheKeyAndForwardedToOrigin:
        EnableAcceptEncodingGzip: true
        EnableAcceptEncodingBrotli: true
        HeadersConfig:
          HeaderBehavior: none
        CookiesConfig:
          CookieBehavior: none
        QueryStringsConfig:
          QueryStringBehavior: none
```

FINOPS SAVINGS — WAF BotControl COMMON Tier Cost vs. TARGETED Tier

- `AWSManagedRulesBotControlRuleSet` at `COMMON` inspection level costs ~$10/month base + per-request charges. This is correct for the current stage. However, `bedrock:*`, `transcribe:*`, `translate:*`, `polly:*` on the IAM role with `Resource: '*'` will generate unbounded AI/ML costs if the Lambda is ever invoked incorrectly or the role is compromised. Scope these down immediately (see Vector 5).

FINOPS SAVINGS — Origin Shield Region: us-east-1 Hardcoded

- Your primary audience spans Europe (ES, PT, FR, IT, DE, PL, SV) and Latin America. Origin Shield in `us-east-1` adds unnecessary inter-region hop latency for European viewers and still bills for Origin Shield requests at $0.0075–$0.012 per 10,000 requests. For a European-heavy language profile, `eu-west-1` (Ireland) or `eu-central-1` (Frankfurt) cuts both latency and cost. Make this a Parameter:

```
Parameters:
  AdminEmail:
    Type: String
    Default: 'info@fixyachips.com'
  OriginShieldRegion:
    Type: String
    Default: 'eu-west-1'
    AllowedValues:
      - 'us-east-1'
      - 'us-west-2'
      - 'eu-west-1'
      - 'eu-central-1'
      - 'ap-southeast-1'
      - 'ap-northeast-1'
    Description: 'Choose closest region to your highest-density viewer base'
```

Then reference in the distribution:

```
OriginShield:
          Enabled: true
          OriginShieldRegion: !Ref OriginShieldRegion
```

VECTOR 2: SYNTAX, STRUCTURAL ERRORS & OMISSIONS

CRITICAL ERROR — ForwardedValues and CachePolicyId Cannot Coexist

- The `DefaultCacheBehavior` uses `ForwardedValues: QueryString: false`. This is the legacy cache configuration method. It **cannot** be used alongside a `CachePolicyId`. You must choose one. AWS will reject the stack update if you attempt to add a CachePolicy while `ForwardedValues` exists. Replace the entire `DefaultCacheBehavior` block:

```
DefaultCacheBehavior:
      TargetOriginId: S3Origin
      ViewerProtocolPolicy: redirect-to-https
      Compress: true
      CachePolicyId: !Ref WebCachePolicy
      OriginRequestPolicyId: !Ref VODOriginRequestPolicy
      ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
      FunctionAssociations:
        - EventType: viewer-request
          FunctionARN: !GetAtt WebsiteRouterFunction.FunctionMetadata.FunctionARN
      AllowedMethods: [GET, HEAD, OPTIONS]
      CachedMethods: [GET, HEAD, OPTIONS]
```

CRITICAL ERROR — Missing CacheBehaviors for VOD Assets (HLS/DASH/MP4)

- There is only a `DefaultCacheBehavior`. There are zero `CacheBehaviors` defined. This means your video segments (`/vod/*`, `/hls/*`, `/dash/*`) are served with the same policy as your HTML pages. This is architecturally broken for VOD delivery:

```
CacheBehaviors:
      # HLS streaming segments - aggressive caching
      - PathPattern: '/hls/*'
        TargetOriginId: S3Origin
        ViewerProtocolPolicy: redirect-to-https
        Compress: false           # Never compress pre-compressed video
        CachePolicyId: !Ref VODCachePolicy
        OriginRequestPolicyId: !Ref VODOriginRequestPolicy
        ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
        AllowedMethods: [GET, HEAD, OPTIONS]
        CachedMethods: [GET, HEAD]
      # DASH streaming segments
      - PathPattern: '/dash/*'
        TargetOriginId: S3Origin
        ViewerProtocolPolicy: redirect-to-https
        Compress: false
        CachePolicyId: !Ref VODCachePolicy
        OriginRequestPolicyId: !Ref VODOriginRequestPolicy
```

```
      ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
      AllowedMethods: [GET, HEAD, OPTIONS]
      CachedMethods: [GET, HEAD]
    # MP4 progressive download
    - PathPattern: '/vod/*'
      TargetOriginId: S3Origin
      ViewerProtocolPolicy: redirect-to-https
      Compress: false
      CachePolicyId: !Ref VODCachePolicy
      OriginRequestPolicyId: !Ref VODOriginRequestPolicy
      ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
      AllowedMethods: [GET, HEAD, OPTIONS]
      CachedMethods: [GET, HEAD]
    # Language-prefixed content paths (ES, PT, FR, IT, DE, PL, SV)
    - PathPattern: '/es/*'
      TargetOriginId: S3Origin
      ViewerProtocolPolicy: redirect-to-https
      Compress: true
      CachePolicyId: !Ref WebCachePolicy
      AllowedMethods: [GET, HEAD]
      CachedMethods: [GET, HEAD]
      FunctionAssociations:
        - EventType: viewer-request
          FunctionARN: !GetAtt WebsiteRouterFunction.FunctionMetadata.FunctionARN
    - PathPattern: '/pt/*'
      TargetOriginId: S3Origin
      ViewerProtocolPolicy: redirect-to-https
      Compress: true
      CachePolicyId: !Ref WebCachePolicy
      AllowedMethods: [GET, HEAD]
      CachedMethods: [GET, HEAD]
      FunctionAssociations:
        - EventType: viewer-request
          FunctionARN: !GetAtt WebsiteRouterFunction.FunctionMetadata.FunctionARN
    - PathPattern: '/fr/*'
      TargetOriginId: S3Origin
      ViewerProtocolPolicy: redirect-to-https
      Compress: true
      CachePolicyId: !Ref WebCachePolicy
      AllowedMethods: [GET, HEAD]
      CachedMethods: [GET, HEAD]
      FunctionAssociations:
        - EventType: viewer-request
          FunctionARN: !GetAtt WebsiteRouterFunction.FunctionMetadata.FunctionARN
    - PathPattern: '/de/*'
      TargetOriginId: S3Origin
      ViewerProtocolPolicy: redirect-to-https
      Compress: true
      CachePolicyId: !Ref WebCachePolicy
      AllowedMethods: [GET, HEAD]
      CachedMethods: [GET, HEAD]
      FunctionAssociations:
        - EventType: viewer-request
          FunctionARN: !GetAtt WebsiteRouterFunction.FunctionMetadata.FunctionARN
    - PathPattern: '/it/*'
      TargetOriginId: S3Origin
      ViewerProtocolPolicy: redirect-to-https
      Compress: true
      CachePolicyId: !Ref WebCachePolicy
```

```
        AllowedMethods: [GET, HEAD]
        CachedMethods: [GET, HEAD]
        FunctionAssociations:
          - EventType: viewer-request
            FunctionARN: !GetAtt WebsiteRouterFunction.FunctionMetadata.FunctionARN
      - PathPattern: '/pl/*'
        TargetOriginId: S3Origin
        ViewerProtocolPolicy: redirect-to-https
        Compress: true
        CachePolicyId: !Ref WebCachePolicy
        AllowedMethods: [GET, HEAD]
        CachedMethods: [GET, HEAD]
        FunctionAssociations:
          - EventType: viewer-request
            FunctionARN: !GetAtt WebsiteRouterFunction.FunctionMetadata.FunctionARN
      - PathPattern: '/sv/*'
        TargetOriginId: S3Origin
        ViewerProtocolPolicy: redirect-to-https
        Compress: true
        CachePolicyId: !Ref WebCachePolicy
        AllowedMethods: [GET, HEAD]
        CachedMethods: [GET, HEAD]
        FunctionAssociations:
          - EventType: viewer-request
            FunctionARN: !GetAtt WebsiteRouterFunction.FunctionMetadata.FunctionARN
```

CRITICAL ERROR — ApiOptionsMethod Uses AWS_PROXY for OPTIONS: Will Fail CORS

- The `ApiOptionsMethod` is wired as `AWS_PROXY` pointing to Lambda. For CORS preflight (`OPTIONS`), the browser never includes a body. The Lambda handler returns OPTIONS correctly via code, BUT `AWS_PROXY` for OPTIONS should use `MOCK` integration with static headers to eliminate a Lambda invocation entirely. Every browser preflight hitting Lambda costs money and adds 50–200ms latency:

```
ApiOptionsMethod:
  Type: 'AWS::ApiGateway::Method'
  Properties:
    AuthorizationType: NONE
    HttpMethod: OPTIONS
    ResourceId: !Ref ApiResource
    RestApiId: !Ref SignupAPI
    Integration:
      Type: MOCK
      RequestTemplates:
        application/json: '{"statusCode": 200}'
      IntegrationResponses:
        - StatusCode: 200
          ResponseParameters:
            method.response.header.Access-Control-Allow-Headers: "'Content-Type,X-
Amz-Date,Authorization,X-Api-Key'"
            method.response.header.Access-Control-Allow-Methods: "'POST,OPTIONS'"
            method.response.header.Access-Control-Allow-Origin:
"'https://fixyachips.com'"
          ResponseTemplates:
            application/json: ''
```

```
    MethodResponses:
      - StatusCode: 200
        ResponseParameters:
          method.response.header.Access-Control-Allow-Headers: false
          method.response.header.Access-Control-Allow-Methods: false
          method.response.header.Access-Control-Allow-Origin: false
```

CRITICAL ERROR — Dead Letter Queue (DLQ) Missing on SignupWorker Lambda

- `SignupWorker` processes waitlist signups. If DynamoDB throttles or SES rejects a send, the event is silently dropped. There is zero DLQ configured. A failed signup during launch day (high concurrency surge) will lose real leads with no recovery path:

```
# Add this SQS queue resource
SignupWorkerDLQ:
  Type: 'AWS::SQS::Queue'
  Properties:
    QueueName: !Sub 'FixyachipsSignupDLQ-${AWS::StackName}'
    MessageRetentionPeriod: 1209600  # 14 days - max retention
    KmsMasterKeyId: alias/aws/sqs    # Encrypt dead letters at rest

# Add DeadLetterConfig to SignupWorker
SignupWorker:
  Type: 'AWS::Lambda::Function'
  Properties:
    # ... existing properties ...
    DeadLetterConfig:
      TargetArn: !GetAtt SignupWorkerDLQ.Arn
```

Also add SQS permissions to the IAM role:

```
- Effect: Allow
          Action:
            - 'sqs:SendMessage'
          Resource: !GetAtt SignupWorkerDLQ.Arn
```

CRITICAL ERROR — CloudFront Distribution Missing Aliases and ViewerCertificate

- `WebsiteDistribution` has no `Aliases` (custom domain) and no `ViewerCertificate`. Without this, the distribution serves from `*.cloudfront.net`, not `fixyachips.com`. Students in 120 countries will see a naked CloudFront URL. Add:

```
Parameters:
  # Add these parameters
  CertificateArn:
    Type: String
    Description: 'ACM Certificate ARN (must be in us-east-1) for fixyachips.com'
    Default: ''
  DomainName:
    Type: String
    Default: 'fixyachips.com'
  WWWDomainName:
    Type: String
```

```
    Default: 'www.fixyachips.com'

DistributionConfig:
      Enabled: true
      PriceClass: PriceClass_All
      DefaultRootObject: index.html
      Aliases:
        - !Ref DomainName
        - !Ref WWWDomainName
      ViewerCertificate:
        AcmCertificateArn: !Ref CertificateArn
        SslSupportMethod: sni-only
        MinimumProtocolVersion: TLSv1.2_2021   # Enforce modern TLS - critical for
TV clients
      WebACLId: !GetAtt WebBodyguard.Arn
```

CRITICAL ERROR — CloudFront Function Runtime cloudfront-js-1.0 is Deprecated

- `Runtime: 'cloudfront-js-1.0'` is the legacy runtime. AWS has released `cloudfront-js-2.0` which supports ES modules, async/await, and has significantly better performance. For an edge function running at 400+ PoPs globally, this matters:

```
FunctionConfig:
      Comment: 'Fixes .html paths for clean URLs and language routing'
      Runtime: 'cloudfront-js-2.0'
    FunctionCode: |
      async function handler(event) {
        const request = event.request;
        const uri = request.uri;
        if (uri.endsWith('/')) {
          request.uri += 'index.html';
        } else if (!uri.includes('.')) {
          request.uri += '.html';
        }
        return request;
      }
```

ARCHITECTURAL ENHANCEMENT — Missing Logging Configuration on CloudFront Distribution

- No `Logging` configuration is defined on `WebsiteDistribution`. Without access logs, you have zero visibility into cache hit ratios per language path, geographic request patterns, or VOD playback failures across your 120 countries:

```
# Add S3 bucket for CloudFront logs
CloudFrontLogsBucket:
  Type: 'AWS::S3::Bucket'
  Properties:
    BucketName: !Sub 'fixyachips-cf-logs-${AWS::AccountId}'
    PublicAccessBlockConfiguration:
      BlockPublicAcls: true
      BlockPublicPolicy: true
      IgnorePublicAcls: true
```

```
      RestrictPublicBuckets: true
    LifecycleConfiguration:
      Rules:
        - Id: ExpireOldLogs
          Status: Enabled
          ExpirationInDays: 90
    OwnershipControls:
      Rules:
        - ObjectOwnership: BucketOwnerPreferred

# Inside DistributionConfig:
      Logging:
        Bucket: !GetAtt CloudFrontLogsBucket.DomainName
        Prefix: 'cloudfront-logs/'
        IncludeCookies: false
```

ARCHITECTURAL ENHANCEMENT — DynamoDB Missing Point-in-Time Recovery (PITR)

- `StudentTable` has `DeletionPolicy: Retain` (good) but zero backup configuration. A corrupted write or accidental `DeleteItem` loop will permanently destroy student records. PITR costs $0.20/GB/month — trivially cheap for a student roster:

```
StudentTable:
  Type: 'AWS::DynamoDB::Table'
  DeletionPolicy: Retain
  UpdateReplacePolicy: Retain
  Properties:
    TableName: !Sub 'FixyachipsStudents-${AWS::StackName}'
    AttributeDefinitions:
      - AttributeName: StudentID
        AttributeType: S
      - AttributeName: Email
        AttributeType: S
    KeySchema:
      - AttributeName: StudentID
        KeyType: HASH
    GlobalSecondaryIndexes:
      - IndexName: EmailIndex
        KeySchema:
          - AttributeName: Email
            KeyType: HASH
        Projection:
          ProjectionType: ALL
    BillingMode: PAY_PER_REQUEST
    PointInTimeRecoverySpecification:
      PointInTimeRecoveryEnabled: true
    SSESpecification:
      SSEEnabled: true
      SSEType: KMS
```

Note the `EmailIndex` GSI added above — without it, looking up a student by email (e.g., to prevent duplicate signups) requires a full `Scan` operation, which is costly and slow at scale.

ARCHITECTURAL ENHANCEMENT — Missing Duplicate Email Check in Lambda

- The `SignupWorker` Lambda does `table.put_item()` directly without checking if the email already exists. A student who double-submits the form will create two `StudentID` records and receive two welcome emails. This is a data integrity failure and an SES reputation risk. The GSI above enables a cheap fix:

```
# Add before table.put_item():
existing = table.query(
    IndexName='EmailIndex',
    KeyConditionExpression='Email = :email',
    ExpressionAttributeValues={':email': email},
    Limit=1
)
if existing.get('Count', 0) > 0:
    return {
        'statusCode': 200,
        'headers': {'Access-Control-Allow-Origin': 'https://fixyachips.com'},
        'body': json.dumps({'status': 'already_registered',
                            'message': 'You are already on the list!'})
    }
```

VECTOR 3: WEB-TO-TV STREAM CONVERSION & VOD DELIVERY

CRITICAL ERROR — No Range Request Support Headers in CloudFront Response Policy

- HTTP Range requests (`Range: bytes=0-`) are the backbone of video scrubbing, fast-forward, and resume-play on every device from Chrome to Samsung TVs to Fire Sticks. CloudFront must explicitly pass `Range` headers to origin AND return the correct response headers. Without this, a Roku or Apple TV client requesting byte range 50MB into a 500MB MP4 will either fail or force a full download restart. Add an Origin Request Policy and Response Headers Policy:

```
VODOriginRequestPolicy:
  Type: 'AWS::CloudFront::OriginRequestPolicy'
  Properties:
    OriginRequestPolicyConfig:
      Name: !Sub 'FixyachipsVODOriginRequest-${AWS::StackName}'
      Comment: 'Forward Range and Origin headers for VOD byte-range serving'
      HeadersConfig:
        HeaderBehavior: whitelist
        Headers:
          - 'Origin'
          - 'Access-Control-Request-Headers'
          - 'Access-Control-Request-Method'
          - 'Range'              # CRITICAL for video scrubbing
      CookiesConfig:
        CookieBehavior: none
      QueryStringsConfig:
```

```
        QueryStringBehavior: none

VODResponseHeadersPolicy:
  Type: 'AWS::CloudFront::ResponseHeadersPolicy'
  Properties:
    ResponseHeadersPolicyConfig:
      Name: !Sub 'FixyachipsVODResponseHeaders-${AWS::StackName}'
      Comment: 'CORS + security headers for VOD delivery'
      CorsConfig:
        AccessControlAllowCredentials: false
        AccessControlAllowHeaders:
          Items: ['*']
        AccessControlAllowMethods:
          Items: [GET, HEAD, OPTIONS]
        AccessControlAllowOrigins:
          Items:
            - 'https://fixyachips.com'
            - 'https://www.fixyachips.com'
        AccessControlExposeHeaders:
          Items:
            - 'Content-Length'
            - 'Content-Range'    # CRITICAL: allows player to track position
            - 'ETag'
            - 'Accept-Ranges'   # CRITICAL: signals byte-range support to players
        AccessControlMaxAgeSec: 3600
        OriginOverride: false
      SecurityHeadersConfig:
        ContentTypeOptions:
          Override: true
        FrameOptions:
          FrameOption: DENY
          Override: true
        ReferrerPolicy:
          ReferrerPolicy: strict-origin-when-cross-origin
          Override: true
        StrictTransportSecurity:
          AccessControlMaxAgeSec: 31536000
          IncludeSubdomains: true
          Preload: true
          Override: true
        XSSProtection:
          ModeBlock: true
          Override: true
          Protection: true
      CustomHeadersConfig:
        Items:
          - Header: 'Cache-Control'
            Value: 'public, max-age=2592000, immutable'
            Override: false
```

ARCHITECTURAL ENHANCEMENT — Missing MediaConvert Integration

- The template claims to be "VOD academy streaming video assets" but there is zero AWS Elemental MediaConvert pipeline defined. Raw MP4 uploads to S3 are not production-ready for global multi-bitrate streaming. You need adaptive bitrate (ABR) encoding to HLS/DASH with

multiple renditions (240p/480p/720p/1080p) so Smart TVs in countries with poor connectivity (parts of Poland, Brazil) auto-adapt:

```
MediaConvertRole:
  Type: 'AWS::IAM::Role'
  Properties:
    RoleName: !Sub 'FixyachipsMediaConvertRole-${AWS::StackName}'
    AssumeRolePolicyDocument:
      Version: '2012-10-17'
      Statement:
        - Effect: Allow
          Principal:
            Service: mediaconvert.amazonaws.com
          Action: 'sts:AssumeRole'
    Policies:
      - PolicyName: MediaConvertS3Access
        PolicyDocument:
          Version: '2012-10-17'
          Statement:
            - Effect: Allow
              Action:
                - 's3:GetObject'
                - 's3:PutObject'
              Resource:
                - !Sub 'arn:aws:s3:::${StreamingVideoBucket}/raw-uploads/*'
                - !Sub 'arn:aws:s3:::${StreamingVideoBucket}/vod/*'
                - !Sub 'arn:aws:s3:::${StreamingVideoBucket}/hls/*'
                - !Sub 'arn:aws:s3:::${StreamingVideoBucket}/dash/*'

VideoProcessingTrigger:
  Type: 'AWS::Lambda::Function'
  Properties:
    FunctionName: !Sub 'FixyachipsVideoProcessor-${AWS::StackName}'
    Handler: index.lambda_handler
    Runtime: python3.12
    Role: !GetAtt FixyachipsMasterRole.Arn
    MemorySize: 512
    Timeout: 60
    Environment:
      Variables:
        MEDIACONVERT_ENDPOINT: ''    # Populate via custom resource or SSM
        MEDIACONVERT_ROLE_ARN: !GetAtt MediaConvertRole.Arn
        OUTPUT_BUCKET: !Ref StreamingVideoBucket
    Code:
      ZipFile: |
        import json, boto3, os, re
        def lambda_handler(event, context):
            mc_client = boto3.client(
                'mediaconvert',
                endpoint_url=os.environ['MEDIACONVERT_ENDPOINT']
            )
            for record in event['Records']:
                key = record['s3']['object']['key']
                if not key.startswith('raw-uploads/'):
                    return
                base_name = re.sub(r'raw-uploads/(.+)\.[^.]+$', r'\1', key)
                job_settings = {
                    'Role': os.environ['MEDIACONVERT_ROLE_ARN'],
                    'Settings': {
```

```
                        'Inputs': [{'FileInput': f"s3://{record['s3']['bucket']
['name']}/{key}"}],
                        'OutputGroups': [
                            {
                                'Name': 'HLS Group',
                                'OutputGroupSettings': {
                                    'Type': 'HLS_GROUP_SETTINGS',
                                    'HlsGroupSettings': {
                                        'Destination':
f"s3://{os.environ['OUTPUT_BUCKET']}/hls/{base_name}/",
                                        'SegmentLength': 6,
                                        'MinSegmentLength': 0
                                    }
                                },
                                'Outputs': [
                                    {'NameModifier': '_1080p', 'VideoDescription':
{'Width': 1920, 'Height': 1080, 'CodecSettings': {'Codec': 'H_264', 'H264Settings':
{'Bitrate': 5000000}}}},
                                    {'NameModifier': '_720p',  'VideoDescription':
{'Width': 1280, 'Height': 720,  'CodecSettings': {'Codec': 'H_264', 'H264Settings':
{'Bitrate': 2500000}}}},
                                    {'NameModifier': '_480p',  'VideoDescription':
{'Width': 854,  'Height': 480,  'CodecSettings': {'Codec': 'H_264', 'H264Settings':
{'Bitrate': 1000000}}}},
                                    {'NameModifier': '_240p',  'VideoDescription':
{'Width': 426,  'Height': 240,  'CodecSettings': {'Codec': 'H_264', 'H264Settings':
{'Bitrate': 400000}}}}
                                ]
                            }
                        ]
                    }
                }
                mc_client.create_job(**job_settings)
                print(f"MediaConvert job submitted for: {key}")

S3VideoUploadNotification:
  Type: 'AWS::S3::BucketNotification'
  # Note: BucketNotification cannot be a separate resource - must be inline in
bucket
  # Use EventBridge instead:

VideoUploadEventRule:
  Type: 'AWS::Events::Rule'
  Properties:
    Name: !Sub 'FixyachipsVideoUpload-${AWS::StackName}'
    EventPattern:
      source: ['aws.s3']
      detail-type: ['Object Created']
      detail:
        bucket:
          name: [!Ref StreamingVideoBucket]
        object:
          key:
            - prefix: 'raw-uploads/'
    Targets:
      - Id: VideoProcessingLambda
        Arn: !GetAtt VideoProcessingTrigger.Arn
```

ARCHITECTURAL ENHANCEMENT — No CloudFront Error Pages for VOD Fallback

- No `CustomErrorResponses` defined. If a student's player requests a VOD segment that doesn't exist (e.g., wrong URL, encoding in progress), CloudFront returns a raw S3 403/404 XML response. Smart TVs and video players cannot parse XML error pages — they freeze or crash. Add graceful error handling:

```
# Inside DistributionConfig:
      CustomErrorResponses:
        - ErrorCode: 403
          ResponseCode: 200
          ResponsePagePath: '/index.html'
          ErrorCachingMinTTL: 10
        - ErrorCode: 404
          ResponseCode: 200
          ResponsePagePath: '/index.html'
          ErrorCachingMinTTL: 10
        - ErrorCode: 500
          ResponseCode: 503
          ResponsePagePath: '/error.html'
          ErrorCachingMinTTL: 5
        - ErrorCode: 503
          ResponseCode: 503
          ResponsePagePath: '/error.html'
          ErrorCachingMinTTL: 5
```

VECTOR 4: MULTI-LINGUAL ARCHITECTURE (8 CORE LANGUAGES)

CRITICAL ERROR — No Language Routing Architecture Exists

- The template has no language routing mechanism whatsoever. The `WebsiteRouterFunction` only handles `.html` suffix rewriting. For 8 languages across 120 countries, there is no cache segmentation, no `Accept-Language` forwarding, and no path-based routing. This is a complete architectural gap. The corrected CloudFront Function below handles clean path routing AND adds language prefix normalization:

```
WebsiteRouterFunction:
  Type: 'AWS::CloudFront::Function'
  Properties:
    Name: !Sub 'FixRouter-${AWS::StackName}'
    AutoPublish: true
    FunctionConfig:
      Comment: 'Clean URLs + Language path normalization for 8-language VOD
academy'
      Runtime: 'cloudfront-js-2.0'
    FunctionCode: |
      const SUPPORTED_LANGUAGES = ['en', 'es', 'pt', 'fr', 'it', 'de', 'pl', 'sv'];
      const DEFAULT_LANGUAGE = 'en';
```

```
      async function handler(event) {
        const request = event.request;
        let uri = request.uri;

        // Extract first path segment to check for language prefix
        const segments = uri.split('/').filter(s => s.length > 0);
        const firstSegment = segments[0] ? segments[0].toLowerCase() : '';

        // If no language prefix, do NOT redirect (kills cache hit ratio)
        // Just serve from /en/ prefix transparently at the S3 level
        if (!SUPPORTED_LANGUAGES.includes(firstSegment)) {
          // Prepend default language for S3 key resolution
          uri = '/' + DEFAULT_LANGUAGE + uri;
        }

        // Clean URL: append index.html for directory paths
        if (uri.endsWith('/')) {
          uri += 'index.html';
        } else if (!uri.includes('.') || uri.split('/').pop().indexOf('.') === -1)
{
          uri += '.html';
        }

        request.uri = uri;
        return request;
      }
```

CRITICAL ERROR — Accept-Language Header Forwarding Will Destroy Cache Hit Ratio

- **Do NOT forward `Accept-Language` headers in the Origin Request Policy for VOD content.** This is the most common mistake in multi-lingual CloudFront architectures. If Accept-Language is included in the cache key, every unique language header variant creates a separate cache entry. A German user on Chrome sends `de-DE,de;q=0.9,en;q=0.8` while a German user on Firefox sends `de,en-US;q=0.7` — these are two different cache keys for the same German `/de/index.html` page, obliterating your cache hit ratio.
- **The correct architecture is path-based routing only**: `/es/*`, `/de/*`, `/fr/*` etc. Content is stored in S3 under language-prefixed paths. CloudFront caches by path. No header forwarding needed. The corrected function above implements this.

ARCHITECTURAL ENHANCEMENT — No Language-Aware S3 Bucket Structure Defined

- The S3 bucket has no enforced folder structure for multi-lingual content. Define your S3 key namespace in documentation and enforce it via IAM path conditions:

```
# Recommended S3 key structure (enforce via CI/CD upload scripts):
# /en/index.html          -> English homepage
# /es/index.html          -> Spanish homepage
# /de/index.html          -> German homepage
# /fr/index.html          -> French homepage
# /it/index.html          -> Italian homepage
# /pt/index.html          -> Portuguese homepage
```

```
# /pl/index.html          -> Polish homepage
# /sv/index.html          -> Swedish homepage
# /hls/{lang}/{course-id}/master.m3u8  -> HLS master playlist per language
# /dash/{lang}/{course-id}/manifest.mpd -> DASH manifest per language
# /vod/{lang}/{course-id}/1080p.mp4    -> Direct MP4 per language
# /raw-uploads/{course-id}.mp4         -> Input for MediaConvert
```

ARCHITECTURAL ENHANCEMENT — No Language Sitemap or hreflang Output

- Missing `CloudFormation Output` for language-specific URLs. Add outputs that your frontend team and SEO tooling can consume:

```
LanguageURLsEN:
    Description: 'English content base URL'
    Value: !Sub 'https://${DomainName}/en/'
  LanguageURLsES:
    Description: 'Spanish content base URL'
    Value: !Sub 'https://${DomainName}/es/'
  LanguageURLsDE:
    Description: 'German content base URL'
    Value: !Sub 'https://${DomainName}/de/'
  LanguageURLsFR:
    Description: 'French content base URL'
    Value: !Sub 'https://${DomainName}/fr/'
  LanguageURLsPT:
    Description: 'Portuguese content base URL'
    Value: !Sub 'https://${DomainName}/pt/'
  LanguageURLsIT:
    Description: 'Italian content base URL'
    Value: !Sub 'https://${DomainName}/it/'
  LanguageURLsPL:
    Description: 'Polish content base URL'
    Value: !Sub 'https://${DomainName}/pl/'
  LanguageURLsSV:
    Description: 'Swedish content base URL'
    Value: !Sub 'https://${DomainName}/sv/'
```

VECTOR 5: SECURITY BLINDSPOTS

CRITICAL ERROR — IAM Role Grants bedrock:*, transcribe:*, translate:*, polly:* with Resource: *

- This is the most dangerous line in the entire template. `'bedrock:*'` on `Resource: '*'` means the `SignupWorker` Lambda (which is internet-exposed via API Gateway) can invoke any Bedrock model, including the most expensive ones like Claude 3 Opus at $15/million tokens. A single malicious request that triggers unexpected code execution could invoke Bedrock in a loop and generate thousands of dollars in charges before you notice. Scope these to specific actions only:

```
# AI/ML: scoped to minimum required actions
        - Effect: Allow
          Action:
            - 'ses:SendEmail'
            - 'ses:SendRawEmail'
          Resource:
            - !Sub 'arn:aws:ses:${AWS::Region}:${AWS::AccountId}:identity/$
{AdminEmail}'
            - !Sub
'arn:aws:ses:${AWS::Region}:${AWS::AccountId}:identity/fixyachips.com'
        # Bedrock: scope to specific models only - NEVER bedrock:*
        - Effect: Allow
          Action:
            - 'bedrock:InvokeModel'
          Resource:
            - !Sub 'arn:aws:bedrock:${AWS::Region}::foundation-model/amazon.titan-
text-express-v1'
              # Add specific model ARNs as needed - never wildcard
        # Translate: scope to translate only
        - Effect: Allow
          Action:
            - 'translate:TranslateText'
            - 'translate:GetTerminology'
          Resource: '*'   # Translate has no resource-level permissions - AWS
limitation
        # Polly: scope to synthesis only
        - Effect: Allow
          Action:
            - 'polly:SynthesizeSpeech'
          Resource: '*'   # Polly has no resource-level permissions - AWS
limitation
        # Transcribe: scope to job management only
        - Effect: Allow
          Action:
            - 'transcribe:StartTranscriptionJob'
            - 'transcribe:GetTranscriptionJob'
          Resource: '*'   # Transcribe has no resource-level permissions - AWS
limitation
```

CRITICAL ERROR — API Gateway POST /signup Has No Authentication and No Rate Limiting

- `AuthorizationType: NONE` on the POST method means any bot on the internet can spam your signup endpoint at full throttle, fill your DynamoDB table with garbage, and exhaust your SES sending quota (getting your domain blacklisted). There is no `UsagePlan`, no `ApiKey`, and no stage-level throttling:

```
# Add API Gateway Stage with throttling
ApiStage:
  Type: 'AWS::ApiGateway::Stage'
  Properties:
    RestApiId: !Ref SignupAPI
    StageName: prod
    MethodSettings:
      - ResourcePath: '/signup'
```

```
        HttpMethod: POST
        ThrottlingRateLimit: 10      # 10 requests per second
        ThrottlingBurstLimit: 50     # 50 burst capacity
      - ResourcePath: '/*'
        HttpMethod: '*'
        ThrottlingRateLimit: 20
        ThrottlingBurstLimit: 100
    AccessLogSetting:
      DestinationArn: !GetAtt ApiLogGroup.Arn
      Format:
'{"requestId":"$context.requestId","ip":"$context.identity.sourceIp","requestTime":
"$context.requestTime","httpMethod":"$context.httpMethod","routeKey":"$context.rout
eKey","status":"$context.status","responseLength":"$context.responseLength"}'

ApiLogGroup:
  Type: 'AWS::Logs::LogGroup'
  Properties:
    LogGroupName: !Sub '/aws/apigateway/FixyachipsAPI-${AWS::StackName}'
    RetentionInDays: 30

# Update ApiDeployment to NOT include StageName (use ApiStage instead)
ApiDeployment:
  Type: 'AWS::ApiGateway::Deployment'
  DependsOn:
    - ApiMethod
    - ApiOptionsMethod
  Properties:
    RestApiId: !Ref SignupAPI
    # Remove StageName from here - managed by ApiStage resource
```

CRITICAL ERROR — WAF Missing Rate-Based Rule

- `WebBodyguard` has `CommonRuleSet` and `BotControlRuleSet` but **no rate-based rule**. A distributed botnet hitting from 10,000 IPs at 1 request/second each will bypass BotControl (which targets known bots) and flood your origin. Add a rate-based rule to catch volumetric attacks:

```
- Name: RateLimitSignupEndpoint
        Priority: 2
        Statement:
          RateBasedStatement:
            Limit: 1000          # Max 1000 requests per 5 minutes per IP
            AggregateKeyType: IP
            ScopeDownStatement:
              ByteMatchStatement:
                SearchString: '/signup'
                FieldToMatch:
                  UriPath: {}
                TextTransformations:
                  - Priority: 0
                    Type: LOWERCASE
                PositionalConstraint: CONTAINS
        Action:
          Block: {}
        VisibilityConfig:
          SampledRequestsEnabled: true
```

```
            CloudWatchMetricsEnabled: true
            MetricName: RateLimitSignup

        - Name: BlockHighRiskCountries
          Priority: 3
          Statement:
            GeoMatchStatement:
              CountryCodes:
                - 'KP'    # North Korea
                - 'IR'    # Iran
                - 'CU'    # Cuba (OFAC)
                - 'SY'    # Syria (OFAC)
          Action:
            Block: {}
          VisibilityConfig:
            SampledRequestsEnabled: true
            CloudWatchMetricsEnabled: true
            MetricName: GeoBlock
```

CRITICAL ERROR — Cognito UserPool Missing MFA Configuration

- `FixyachipsUserPool` has no MFA settings. For a paid VOD academy with paywall-ready architecture, account takeover via credential stuffing is the #1 identity threat. Without MFA, a compromised password = full account access to paid content:

```
FixyachipsUserPool:
  Type: 'AWS::Cognito::UserPool'
  Properties:
    UserPoolName: !Sub 'Fixyachips_Global_Users-${AWS::StackName}'
    UsernameAttributes: [email]
    AutoVerifiedAttributes: [email]
    MfaConfiguration: OPTIONAL     # Don't force MFA at launch - kills conversion
    EnabledMfas:
      - TOTP                       # Google Authenticator / Authy
    AccountRecoverySetting:
      RecoveryMechanisms:
        - Name: verified_email
          Priority: 1
    AdminCreateUserConfig:
      AllowAdminCreateUserOnly: false
      InviteMessageTemplate:
        EmailMessage: 'Welcome to Fixyachips Academy. Your temporary password is
{####}'
        EmailSubject: 'Your Fixyachips Academy Access'
    UserPoolAddOns:
      AdvancedSecurityMode: ENFORCED   # Blocks compromised credentials
    Policies:
      PasswordPolicy:
        MinimumLength: 8
        RequireLowercase: true
        RequireNumbers: true
        RequireSymbols: true
        RequireUppercase: true
        TemporaryPasswordValidityDays: 7
```

CRITICAL ERROR — Cognito Client Allows ALLOW_USER_PASSWORD_AUTH

- `ALLOW_USER_PASSWORD_AUTH` sends the plaintext password to Cognito. This auth flow is vulnerable to MITM if TLS is ever misconfigured and is the weakest auth flow available. For a public-facing SPA, use `ALLOW_USER_SRP_AUTH` exclusively (Secure Remote Password — password never leaves the client in plaintext):

```
FixyachipsUserPoolClient:
  Type: 'AWS::Cognito::UserPoolClient'
  Properties:
    ClientName: Fixyachips_Web_Portal
    UserPoolId: !Ref FixyachipsUserPool
    GenerateSecret: false
    ExplicitAuthFlows:
      - ALLOW_USER_SRP_AUTH          # KEEP: secure password auth
      - ALLOW_REFRESH_TOKEN_AUTH     # KEEP: token refresh
      # REMOVED: ALLOW_USER_PASSWORD_AUTH - plaintext password risk
    PreventUserExistenceErrors: ENABLED   # Don't leak whether email exists
    RefreshTokenValidity: 30
    AccessTokenValidity: 1
    IdTokenValidity: 1
    TokenValidityUnits:
      RefreshToken: days
      AccessToken: hours
      IdToken: hours
    EnableTokenRevocation: true
```

ARCHITECTURAL ENHANCEMENT — Missing CloudFront Geographic Restriction for Regional Licensing

- The template has zero `Restrictions` block. If fixyachips.com ever acquires regional licensing restrictions (common in VOD/education platforms), you have no mechanism to enforce them. Add the structure now even with an empty allowlist so it's ready to activate:

```
# Inside DistributionConfig:
      Restrictions:
        GeoRestriction:
          RestrictionType: none    # Change to 'blacklist' or 'whitelist' when
licensing requires
          # Locations: []          # Uncomment and populate when needed
          # Example blacklist: ['KP', 'IR', 'CU', 'SY']
```

ARCHITECTURAL ENHANCEMENT — Missing S3 Bucket Versioning and Replication for VOD Assets

- If a video file is accidentally overwritten or deleted in S3, there is no recovery path. Enable versioning and consider cross-region replication for disaster recovery of master video assets:

```
StreamingVideoBucket:
  # Add to Properties:
    VersioningConfiguration:
      Status: Enabled
```

ARCHITECTURAL ENHANCEMENT — Missing CloudWatch Alarms for Security Events

- Zero operational alarms defined anywhere in the template. For a global platform you need at minimum:

```
WAFBlockedRequestsAlarm:
  Type: 'AWS::CloudWatch::Alarm'
  Properties:
    AlarmName: !Sub 'FixyachipsWAFBlocks-${AWS::StackName}'
    AlarmDescription: 'WAF blocking abnormal volume of requests - potential attack'
    MetricName: BlockedRequests
    Namespace: AWS/WAFV2
    Dimensions:
      - Name: WebACL
        Value: !Sub 'FixyachipsShield-${AWS::StackName}'
      - Name: Region
        Value: 'us-east-1'
      - Name: Rule
        Value: ALL
    Statistic: Sum
    Period: 300
    EvaluationPeriods: 1
    Threshold: 1000
    ComparisonOperator: GreaterThanThreshold
    AlarmActions:
      - !Ref SecurityAlertTopic

SignupAnomalyAlarm:
  Type: 'AWS::CloudWatch::Alarm'
  Properties:
    AlarmName: !Sub 'FixyachipsSignupAnomaly-${AWS::StackName}'
    AlarmDescription: 'Unusual signup volume - possible bot or viral event'
    MetricName: Invocations
    Namespace: AWS/Lambda
    Dimensions:
      - Name: FunctionName
        Value: !Ref SignupWorker
    Statistic: Sum
    Period: 300
    EvaluationPeriods: 1
    Threshold: 500
    ComparisonOperator: GreaterThanThreshold
    AlarmActions:
      - !Ref SecurityAlertTopic

SecurityAlertTopic:
  Type: 'AWS::SNS::Topic'
  Properties:
    TopicName: !Sub 'FixyachipsSecurityAlerts-${AWS::StackName}'
    Subscription:
      - Protocol: email
        Endpoint: !Ref AdminEmail
```

COMPLETE CORRECTED DISTRIBUTION BLOCK (PRODUCTION-READY)

The following is the fully corrected `WebsiteDistribution` resource incorporating all fixes from Vectors 1–5:

```
WebsiteDistribution:
  Type: 'AWS::CloudFront::Distribution'
  DependsOn:
    - EmpireOAC
    - WebBodyguard
    - VODCachePolicy
    - WebCachePolicy
    - VODOriginRequestPolicy
    - VODResponseHeadersPolicy
  Properties:
    DistributionConfig:
      Enabled: true
      PriceClass: PriceClass_All
      DefaultRootObject: index.html
      Aliases:
        - !Ref DomainName
        - !Ref WWWDomainName
      ViewerCertificate:
        AcmCertificateArn: !Ref CertificateArn
        SslSupportMethod: sni-only
        MinimumProtocolVersion: TLSv1.2_2021
      WebACLId: !GetAtt WebBodyguard.Arn
      HttpVersion: http2and3          # Enable HTTP/3 for modern TV clients
      IPV6Enabled: true
      Logging:
        Bucket: !GetAtt CloudFrontLogsBucket.DomainName
        Prefix: 'cloudfront-logs/'
        IncludeCookies: false
      Origins:
        - Id: S3Origin
          DomainName: !GetAtt StreamingVideoBucket.RegionalDomainName
          S3OriginConfig:
            OriginAccessIdentity: ""
          OriginAccessControlId: !GetAtt EmpireOAC.Id
          OriginShield:
            Enabled: true
            OriginShieldRegion: !Ref OriginShieldRegion
      DefaultCacheBehavior:
        TargetOriginId: S3Origin
        ViewerProtocolPolicy: redirect-to-https
        Compress: true
        CachePolicyId: !Ref WebCachePolicy
        ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
        AllowedMethods: [GET, HEAD, OPTIONS]
        CachedMethods: [GET, HEAD, OPTIONS]
        FunctionAssociations:
          - EventType: viewer-request
            FunctionARN: !GetAtt WebsiteRouterFunction.FunctionMetadata.FunctionARN
      CacheBehaviors:
        - PathPattern: '/hls/*'
          TargetOriginId: S3Origin
          ViewerProtocolPolicy: redirect-to-https
```

```
          Compress: false
          CachePolicyId: !Ref VODCachePolicy
          OriginRequestPolicyId: !Ref VODOriginRequestPolicy
          ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
          AllowedMethods: [GET, HEAD, OPTIONS]
          CachedMethods: [GET, HEAD]
        - PathPattern: '/dash/*'
          TargetOriginId: S3Origin
          ViewerProtocolPolicy: redirect-to-https
          Compress: false
          CachePolicyId: !Ref VODCachePolicy
          OriginRequestPolicyId: !Ref VODOriginRequestPolicy
          ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
          AllowedMethods: [GET, HEAD, OPTIONS]
          CachedMethods: [GET, HEAD]
        - PathPattern: '/vod/*'
          TargetOriginId: S3Origin
          ViewerProtocolPolicy: redirect-to-https
          Compress: false
          CachePolicyId: !Ref VODCachePolicy
          OriginRequestPolicyId: !Ref VODOriginRequestPolicy
          ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
          AllowedMethods: [GET, HEAD, OPTIONS]
          CachedMethods: [GET, HEAD]
        - PathPattern: '/es/*'
          TargetOriginId: S3Origin
          ViewerProtocolPolicy: redirect-to-https
          Compress: true
          CachePolicyId: !Ref WebCachePolicy
          ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
          AllowedMethods: [GET, HEAD]
          CachedMethods: [GET, HEAD]
          FunctionAssociations:
            - EventType: viewer-request
              FunctionARN: !GetAtt
WebsiteRouterFunction.FunctionMetadata.FunctionARN
        - PathPattern: '/pt/*'
          TargetOriginId: S3Origin
          ViewerProtocolPolicy: redirect-to-https
          Compress: true
          CachePolicyId: !Ref WebCachePolicy
          ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
          AllowedMethods: [GET, HEAD]
          CachedMethods: [GET, HEAD]
          FunctionAssociations:
            - EventType: viewer-request
              FunctionARN: !GetAtt
WebsiteRouterFunction.FunctionMetadata.FunctionARN
        - PathPattern: '/fr/*'
          TargetOriginId: S3Origin
          ViewerProtocolPolicy: redirect-to-https
          Compress: true
          CachePolicyId: !Ref WebCachePolicy
          ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
          AllowedMethods: [GET, HEAD]
          CachedMethods: [GET, HEAD]
          FunctionAssociations:
            - EventType: viewer-request
```

```
              FunctionARN: !GetAtt
WebsiteRouterFunction.FunctionMetadata.FunctionARN
        - PathPattern: '/it/*'
          TargetOriginId: S3Origin
          ViewerProtocolPolicy: redirect-to-https
          Compress: true
          CachePolicyId: !Ref WebCachePolicy
          ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
          AllowedMethods: [GET, HEAD]
          CachedMethods: [GET, HEAD]
          FunctionAssociations:
            - EventType: viewer-request
              FunctionARN: !GetAtt
WebsiteRouterFunction.FunctionMetadata.FunctionARN
        - PathPattern: '/de/*'
          TargetOriginId: S3Origin
          ViewerProtocolPolicy: redirect-to-https
          Compress: true
          CachePolicyId: !Ref WebCachePolicy
          ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
          AllowedMethods: [GET, HEAD]
          CachedMethods: [GET, HEAD]
          FunctionAssociations:
            - EventType: viewer-request
              FunctionARN: !GetAtt
WebsiteRouterFunction.FunctionMetadata.FunctionARN
        - PathPattern: '/pl/*'
          TargetOriginId: S3Origin
          ViewerProtocolPolicy: redirect-to-https
          Compress: true
          CachePolicyId: !Ref WebCachePolicy
          ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
          AllowedMethods: [GET, HEAD]
          CachedMethods: [GET, HEAD]
          FunctionAssociations:
            - EventType: viewer-request
              FunctionARN: !GetAtt
WebsiteRouterFunction.FunctionMetadata.FunctionARN
        - PathPattern: '/sv/*'
          TargetOriginId: S3Origin
          ViewerProtocolPolicy: redirect-to-https
          Compress: true
          CachePolicyId: !Ref WebCachePolicy
          ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
          AllowedMethods: [GET, HEAD]
          CachedMethods: [GET, HEAD]
          FunctionAssociations:
            - EventType: viewer-request
              FunctionARN: !GetAtt
WebsiteRouterFunction.FunctionMetadata.FunctionARN
      CustomErrorResponses:
        - ErrorCode: 403
          ResponseCode: 200
          ResponsePagePath: '/index.html'
          ErrorCachingMinTTL: 10
        - ErrorCode: 404
          ResponseCode: 200
          ResponsePagePath: '/index.html'
          ErrorCachingMinTTL: 10
```

```
    - ErrorCode: 500
      ResponseCode: 503
      ResponsePagePath: '/en/error.html'
      ErrorCachingMinTTL: 5
  Restrictions:
    GeoRestriction:
      RestrictionType: none
```

SEVERITY SUMMARY TABLE

Finding	Severity	Vector
`bedrock:*` wildcard on internet-exposed Lambda role	CRITICAL ERROR	5
No ACM certificate / custom domain on CloudFront	CRITICAL ERROR	2
ForwardedValues blocks cache policy adoption	CRITICAL ERROR	2
No Range request headers — video scrubbing broken on TVs	CRITICAL ERROR	3
OPTIONS method using Lambda proxy instead of MOCK	CRITICAL ERROR	2
No DLQ on SignupWorker — leads lost on surge	CRITICAL ERROR	2
No rate limiting on /signup API Gateway method	CRITICAL ERROR	5
ALLOW_USER_PASSWORD_AUTH on Cognito client	CRITICAL ERROR	5
No language routing architecture (8 languages, zero config)	CRITICAL ERROR	4
No VOD CacheBehaviors — video served with HTML TTLs	CRITICAL ERROR	3
CloudFront Function on deprecated JS 1.0 runtime	CRITICAL ERROR	2
No WAF rate-based rule — volumetric attacks bypass WAF	CRITICAL ERROR	5
S3 Lifecycle/Intelligent-Tiering absent on VOD assets	FINOPS SAVINGS	1
Lambda memory unset — 128MB cold start bleed	FINOPS SAVINGS	1
Lambda timeout 30s — 2x correct value, billing waste	FINOPS SAVINGS	1
Origin Shield hardcoded us-east-1 for European audience	FINOPS SAVINGS	1
Missing TTL config — CloudFront falls back to origin headers	FINOPS SAVINGS	1
S3 CORS allows POST/PUT on read-only CDN bucket	FINOPS SAVINGS	1
No MediaConvert pipeline — raw MP4 not ABR-encoded	ARCHITECTURAL	3
No CloudFront access logging	ARCHITECTURAL	2
DynamoDB missing PITR and Email GSI	ARCHITECTURAL	2
Cognito missing MFA and AdvancedSecurity	ARCHITECTURAL	5
No CloudWatch alarms for WAF/Lambda anomalies	ARCHITECTURAL	5
No S3 bucket versioning for VOD master assets	ARCHITECTURAL	5
No geo-restriction structure for licensing readiness	ARCHITECTURAL	5
HTTP/3 not enabled on CloudFront (Smart TV benefit)	ARCHITECTURAL	3
Duplicate email check missing in Lambda	ARCHITECTURAL	2



is that the entire yaml file?

No — that is NOT the entire YAML file. The PDF you uploaded was **cut off**. Here is exactly what was captured vs. what is missing:

What Was Successfully Captured (17 Resources)

The PDF contained the complete definitions for all 17 resources listed in the file header:

1. `EmpireOAC` ✅
2. `WebBodyguard` ✅
3. `WebsiteRouterFunction` ✅
4. `StreamingVideoBucket` ✅
5. `BucketPolicy` ✅
6. `WebsiteDistribution` ✅
7. `StudentTable` ✅
8. `FixyachipsMasterRole` ✅
9. `SignupWorker` ✅ (Lambda + inline Python code)
10. `SignupWorkerLogGroup` ✅
11. `SignupAPI` ✅
12. `ApiResource` ✅
13. `ApiMethod` ✅
14. `ApiDeployment` ✅
15. `LambdaPermission` ✅
16. `FixyachipsUserPool` ✅
17. `FixyachipsUserPoolClient` ✅
18. `Outputs` block ✅

What Concerns Me About Completeness

The description says **"v4.0 COMPLETE STACK — OAC, Origin Shield, WAF, AI Suite, Signup API, Paywall-Ready"** — but based on what was audited, the following **described capabilities have zero corresponding resources** in the file:

Claimed Feature	Resources Present	Verdict
AI Suite	Only IAM permissions for Bedrock/Translate/Polly	**No AI Lambda functions defined**
Paywall-Ready	Only Cognito UserPool exists	**No payment/entitlement resources**
VOD Streaming	Only S3 bucket exists	**No MediaConvert, no HLS**

Claimed Feature	Resources Present	Verdict
Media Pipeline	Nothing	**Completely absent pipeline**

My Recommendation

Please paste the raw YAML text directly into the chat rather than uploading as PDF. PDFs truncate code blocks, collapse indentation, and wrap long lines — all of which are critical to accurate CloudFormation auditing.

If you paste the raw YAML I can:

- Confirm whether the AI Suite functions exist but were cut from the PDF
- Audit the paywall/entitlement architecture
- Check if there are additional Lambda functions, Step Functions, EventBridge rules, or MediaConvert jobs that were not visible in the upload
- Give you a fully complete audit with zero blind spots from document truncation

The audit I delivered is valid for what was visible — but you may have additional resources in the actual file that were never reviewed.



lets start over here is a cAWSTemplateFormatVersion: '2010-09-09' Description: 'Fixyachips Global v4.0: COMPLETE STACK - OAC, Origin Shield, WAF, AI Suite, Signup API, Paywall-Ready'

==========

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WHAT'S IN THIS FILE (v4.0 COMPLETE)

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1. EmpireOAC - Origin Access Control (S3 lock)

2. WebBodyguard - WAF firewall (restored from v3.4)

3. WebsiteRouterFunction - CloudFront edge URL router

4. StreamingVideoBucket - Secured S3 bucket (CORS locked)

5. BucketPolicy - Correct ARN handshake (FIXED)

6. WebsiteDistribution - CloudFront + Origin Shield + WAF

7. StudentTable - DynamoDB (DeletionPolicy: Retain)

8. FixyachipsMasterRole - Scoped IAM (FIXED - not wildcard)

9. SignupWorker - Lambda pre-signup + SES email

10. SignupWorkerLogGroup - CloudWatch logs for Lambda

11. SignupAPI - API Gateway REST API

12. ApiResource - /signup route

13. ApiMethod - POST handler

14. ApiDeployment - Deploys API to 'prod' stage

15. LambdaPermission - Lets API Gateway invoke Lambda

16. FixyachipsUserPool - Cognito user pool

17. FixyachipsUserPoolClient - Cognito web client

==========

Parameters: AdminEmail: Type: String Default: 'info@fixyachips.com'

Resources:

==========

==========

1. THE LOCK (Origin Access Control)

Tells CloudFront to sign every S3 request with SigV4.

This means your S3 bucket stays 100% private.

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==========

EmpireOAC: Type: 'AWS::CloudFront::OriginAccessControl' Properties: OriginAccessControlConfig: Name: !Sub 'Empire-OAC-${AWS::StackName}' OriginAccessControlOriginType: s3 SigningBehavior: always SigningProtocol: sigv4

==========

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2. THE BODYGUARD (WAF - Web Application Firewall)

Restored from v3.4. Protects against bots, SQL injection,

XSS attacks, and common exploits at the CloudFront edge.

Cost: ~5/monthbase+5/month base + 5/monthbase+0.60 per million requests.

MUST be in us-east-1 for CloudFront scope.

==========

==========

WebBodyguard: Type: 'AWS::WAFv2::WebACL' Properties: Name: !Sub 'FixyachipsShield-${AWS::StackName}' Scope: CLOUDFRONT DefaultAction: { Allow: {} } VisibilityConfig: SampledRequestsEnabled: true CloudWatchMetricsEnabled: true MetricName: FixyachipsWAF Rules: - Name: AWS-AWSManagedRulesCommonRuleSet Priority: 0 Statement: ManagedRuleGroupStatement: VendorName: AWS Name: AWSManagedRulesCommonRuleSet OverrideAction: { None: {} } VisibilityConfig: SampledRequestsEnabled: true CloudWatchMetricsEnabled: true MetricName: CommonRules - Name: AWS-AWSManagedRulesBotControlRuleSet Priority: 1 Statement: ManagedRuleGroupStatement: VendorName: AWS Name: AWSManagedRulesBotControlRuleSet ManagedRuleGroupConfigs: - AWSManagedRulesBotControlRuleSet: InspectionLevel: COMMON OverrideAction: { None: {} } VisibilityConfig: SampledRequestsEnabled: true CloudWatchMetricsEnabled: true MetricName: BotControl

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3. THE EDGE ROUTER (CloudFront Function)

Handles clean URLs - turns /about into /about.html

Runs at the edge in every CloudFront location worldwide.

==========

==========

WebsiteRouterFunction: Type: 'AWS::CloudFront::Function' Properties: Name: !Sub 'FixRouter-${AWS::StackName}' AutoPublish: true FunctionConfig: Comment: 'Fixes .html paths for clean URLs' Runtime: 'cloudfront-js-1.0' FunctionCode: | function handler(event) { var request = event.request; var uri = request.uri; if (uri.endsWith('/')) { request.uri += 'index.html'; } else if (!uri.includes('.')) { request.uri += '.html'; } return request; }

==========

==========

4. THE STORAGE (S3 Bucket)

Fully private - no public access whatsoever.

CORS locked to fixyachips.com only (FIXED from v3.9).

==========

==========

StreamingVideoBucket: Type: 'AWS::S3::Bucket' Properties: BucketName: !Sub 'fixyachips-platform-${AWS::AccountId}' PublicAccessBlockConfiguration: BlockPublicAcls: true BlockPublicPolicy: true IgnorePublicAcls: true RestrictPublicBuckets: true CorsConfiguration: CorsRules: - AllowedHeaders: ['*'] AllowedMethods: [GET, POST, PUT] AllowedOrigins: - 'https://fixyachips.com' - 'https://www.fixyachips.com'

==========

==========

5. THE HANDSHAKE (Bucket Policy)

Only allows CloudFront (via OAC) to read from S3.

ARN fixed from v3.9 (was 'arn:aws:partition' - not real).

==========

==========

BucketPolicy: Type: 'AWS::S3::BucketPolicy' Properties: Bucket: !Ref StreamingVideoBucket PolicyDocument: Version: '2012-10-17' Statement: - Effect: Allow Principal: Service: cloudfront.amazonaws.com Action: 's3:GetObject' Resource: !Sub 'arn:aws:s3:::StreamingVideoBucket/ * 'Condition:StringEquals:AWS:SourceArn:!Sub 'arn:aws:cloudfront::{StreamingVideoBucket}/*' Condition: StringEquals: AWS:SourceArn: !Sub 'arn:aws:cloudfront::StreamingVideoBucket/ * 'Condition:StringEquals:AWS:SourceArn:!Sub 'arn:aws:cloudfront::{AWS::AccountId}:distribution/${WebsiteDistribution}'

==========

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6. THE GLOBAL WEBSITE (CloudFront Distribution)

PriceClass_All = all 400+ edge locations worldwide.

Origin Shield = extra caching layer, saves S3 costs.

WAF wired in via WebAclId.

==========

==========

WebsiteDistribution: Type: 'AWS::CloudFront::Distribution' Properties: DistributionConfig: Enabled: true PriceClass: PriceClass_All DefaultRootObject: index.html # WAF connected to CloudFront distribution WebACLId: !GetAtt WebBodyguard.Arn Origins: - Id: S3Origin DomainName: !GetAtt StreamingVideoBucket.RegionalDomainName # S3OriginConfig required as placeholder even when using OAC S3OriginConfig: OriginAccessIdentity: "" OriginAccessControlId: !GetAtt EmpireOAC.Id # Origin Shield: extra caching layer between CloudFront and S3 OriginShield: Enabled: true OriginShieldRegion: us-east-1 DefaultCacheBehavior: TargetOriginId: S3Origin ViewerProtocolPolicy: redirect-to-https Compress: true ForwardedValues: QueryString: false FunctionAssociations: - EventType: viewer-request FunctionARN: !GetAtt WebsiteRouterFunction.FunctionMetadata.FunctionARN

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7. THE DATABASE (DynamoDB)

Stores every student record.

DeletionPolicy: Retain = data survives even if stack deleted.

PAY_PER_REQUEST = $0 cost until you have real students.

==========

==========

StudentTable: Type: 'AWS::DynamoDB::Table' DeletionPolicy: Retain Properties: TableName: !Sub 'FixyachipsStudents-${AWS::StackName}' AttributeDefinitions: - AttributeName: StudentID AttributeType: S KeySchema: - AttributeName: StudentID KeyType: HASH BillingMode: PAY_PER_REQUEST

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8. THE BRAINS (IAM Role)

Used by all Lambda functions.

FIXED: Scoped to specific resources only (not wildcard *).

==========

==========

FixyachipsMasterRole: Type: 'AWS::IAM::Role' Properties: AssumeRolePolicyDocument: Version: '2012-10-17' Statement: - Effect: Allow Principal: { Service: [lambda.amazonaws.com] } Action: ['sts:AssumeRole'] ManagedPolicyArns: - 'arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole' Policies: - PolicyName: FixyachipsAIPowerhouse PolicyDocument: Version: '2012-10-17' Statement: # S3: scoped to your bucket only - Effect: Allow Action: - 's3:GetObject' - 's3:PutObject' - 's3:DeleteObject' - 's3:ListBucket' Resource: - !Sub 'arn:aws:s3:::StreamingVideoBucket′−!Sub′arn:aws:s3:::{StreamingVideoBucket}' - ! Sub 'arn:aws:s3:::StreamingVideoBucket′−!Sub′arn:aws:s3:::{StreamingVideoBucket}/' *# DynamoDB: scoped to your student table only - Effect: Allow Action: - 'dynamodb:GetItem' - 'dynamodb:PutItem' - 'dynamodb:UpdateItem' - 'dynamodb:DeleteItem' - 'dynamodb:Query' - 'dynamodb:Scan' Resource: ! GetAtt StudentTable.Arn # AI/ML: these services require * resource (AWS limitation) - Effect: Allow Action: - 'ses:SendEmail' - 'bedrock:*'* - 'transcribe:' - *'translate:*' - 'polly:' *Resource: "*

==========

==========

9. THE SIGNUP WORKER (Lambda Function)

Restored from v3.4. Handles pre-launch waitlist signups.

Flow: Frontend form -> API Gateway -> This Lambda ->

DynamoDB (stores lead) + SES (sends welcome email)

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==========

SignupWorker: Type: 'AWS::Lambda::Function' Properties: FunctionName: !Sub 'FixyachipsSignupWorker-${AWS::StackName}' Handler: index.lambda_handler Runtime: python3.12 Role: !GetAtt FixyachipsMasterRole.Arn Timeout: 30 Environment: Variables: TABLE_NAME: !Ref StudentTable ADMIN_EMAIL: !Ref AdminEmail Code: ZipFile: | import json, boto3, uuid, os dynamodb = boto3.resource('dynamodb') ses = boto3.client('ses') table = dynamodb.Table(os.environ['TABLE_NAME'])

```
def lambda_handler(event, context):

          if event.get('httpMethod') == 'OPTIONS':
              return {
                  'statusCode': 200,
                  'headers': {
                      'Access-Control-Allow-Origin': 'https://fixyachips.com',
                      'Access-Control-Allow-Headers': 'Content-Type',
                      'Access-Control-Allow-Methods': 'POST, OPTIONS'
                  },
                  'body': ''
              }
          try:
              body = json.loads(event['body'])
              email = body.get('email', '').strip().lower()
```

```
            name = body.get('name', '').strip()

            if not email or not name:
                return {
                    'statusCode': 400,
                    'headers': {'Access-Control-Allow-Origin':
'https://fixyachips.com'},
                    'body': json.dumps({'error': 'Name and email are required'})
                }


            table.put_item(Item={
                'StudentID': str(uuid.uuid4()),
                'Name': name,
                'Email': email,
                'Status': 'WAITLIST',
                'Source': body.get('source', 'website'),
                'Industry': body.get('industry', 'unknown')
            })


            ses.send_email(
                Source=os.environ['ADMIN_EMAIL'],
                Destination={'ToAddresses': [email]},
                Message={
                    'Subject': {'Data': 'Welcome to the No-Drill Revolution -
Fixyachips Academy'},
                    'Body': {'Text': {'Data': 'Hello ' + name + ',\n\nYou are now
on the waitlist for Fixyachips Global Academy.\n\nYou are about to learn the same
trade secret No-Drill windshield repair methodology passed down since 1972 -
personally taught to Joshua Cane by the pioneers themselves.\n\nThis is not just a
course. This is a business in a box.\n\nWe will notify you the moment enrollment
opens.\n\nJoshua Cane\nFounder, Fixyachips Global Academy\nCustodian of the 1972
No-Drill Methodology'}}
                }
            )

            ses.send_email(
                Source=os.environ['ADMIN_EMAIL'],
                Destination={'ToAddresses': [os.environ['ADMIN_EMAIL']]},
                Message={
                    'Subject': {'Data': f'New Waitlist Signup: {name}'},
                    'Body': {'Text': {'Data': f'New signup:\nName: {name}\nEmail:
{email}\nIndustry: {body.get("industry", "unknown")}'}}
                }
            )

            return {
                'statusCode': 200,
                'headers': {
                    'Access-Control-Allow-Origin': 'https://fixyachips.com',
                    'Content-Type': 'application/json'
                },
                'body': json.dumps({'status': 'success', 'message': 'Welcome to
the revolution!'})
            }

        except Exception as e:
            print(f"ERROR: {str(e)}")
```

```
            return {
                'statusCode': 500,
                'headers': {'Access-Control-Allow-Origin':
'https://fixyachips.com'},
                'body': json.dumps({'error': 'Signup failed. Please try again.'})
            }
```

==========

==========

10. CLOUDWATCH LOGS (Lambda Log Group)

Gives your Lambda a named log group with 30-day retention.

Without this logs pile up forever and cost money.

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==========

SignupWorkerLogGroup: Type: 'AWS::Logs::LogGroup' Properties: LogGroupName: !Sub '/aws/lambda/FixyachipsSignupWorker-${AWS::StackName}' RetentionInDays: 30

==========

==========

11. THE FRONT DOOR (API Gateway)

Creates the REST API that your frontend form posts to.

==========

==========

SignupAPI: Type: 'AWS::ApiGateway::RestApi' Properties: Name: !Sub 'FixyachipsAPI-${AWS::StackName}' Description: 'Fixyachips Signup and Course API'

==========

==========

12. THE ROUTE (/signup resource)

==========

==========

ApiResource: Type: 'AWS::ApiGateway::Resource' Properties: ParentId: !GetAtt SignupAPI.RootResourceId PathPart: 'signup' RestApiId: !Ref SignupAPI

==========

==========

13. THE METHOD (POST handler)

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==========

ApiMethod: Type: 'AWS::ApiGateway::Method' Properties: AuthorizationType: NONE HttpMethod: POST ResourceId: !Ref ApiResource RestApiId: !Ref SignupAPI Integration: IntegrationHttpMethod: POST Type: AWS_PROXY Uri: !Sub 'arn:aws:apigateway:AWS::Region:lambda:path/2015−03−31/functions/{AWS::Region}:lambda:path/2015-03-31/functions/AWS::Region:lambda:path/2015−03−31/functions/{SignupWorker.Arn}/invocations'

CORS OPTIONS method for browser preflight requests

ApiOptionsMethod: Type: 'AWS::ApiGateway::Method' Properties: AuthorizationType: NONE HttpMethod: OPTIONS ResourceId: !Ref ApiResource RestApiId: !Ref SignupAPI Integration: IntegrationHttpMethod: POST Type: AWS_PROXY Uri: !Sub 'arn:aws:apigateway:AWS::Region:lambda:path/2015−03−31/functions/{AWS::Region}:lambda:path/2015-03-31/functions/AWS::Region:lambda:path/2015−03−31/functions/{SignupWorker.Arn}/invocations'

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14. THE DEPLOYMENT (publishes API to 'prod' stage)

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==========

ApiDeployment: Type: 'AWS::ApiGateway::Deployment' DependsOn: - ApiMethod - ApiOptionsMethod Properties: RestApiId: !Ref SignupAPI StageName: prod

==========

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15. LAMBDA PERMISSION

Lets API Gateway invoke the SignupWorker Lambda.

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==========

LambdaPermission: Type: 'AWS::Lambda::Permission' Properties: Action: 'lambda:InvokeFunction' FunctionName: !Ref SignupWorker Principal: apigateway.amazonaws.com SourceArn: !Sub 'arn:aws:execute-api:AWS::Region:{AWS::Region}:AWS::Region:{AWS::AccountId}:${SignupAPI}/*'

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16. COGNITO USER POOL

Handles student login/signup with email verification.

Custom attributes: veteran_status, industry_segment.

Strong password policy from v3.4 restored.

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FixyachipsUserPool: Type: 'AWS::Cognito::UserPool' Properties: UserPoolName: !Sub 'Fixyachips_Global_Users-${AWS::StackName}' UsernameAttributes: [email] AutoVerifiedAttributes: [email] Policies: PasswordPolicy: MinimumLength: 8 RequireLowercase: true RequireNumbers: true RequireSymbols: true RequireUppercase: true Schema: - Name: email AttributeDataType: String Mutable: true Required: true - Name: veteran_status AttributeDataType: String Mutable: true - Name: industry_segment AttributeDataType: String Mutable: true

==========

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17. COGNITO USER POOL CLIENT

The web app client - no secret (public SPA client).

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==========

FixyachipsUserPoolClient: Type: 'AWS::Cognito::UserPoolClient' Properties: ClientName: Fixyachips_Web_Portal UserPoolId: !Ref FixyachipsUserPool GenerateSecret: false ExplicitAuthFlows: - ALLOW_USER_PASSWORD_AUTH - ALLOW_REFRESH_TOKEN_AUTH - ALLOW_USER_SRP_AUTH

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OUTPUTS - Values you'll need for your frontend code

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Outputs: WebsiteURL: Description: 'Your CloudFront website URL' Value: !GetAtt WebsiteDistribution.DomainName

SignupApiUrl: Description: 'POST endpoint for waitlist signups' Value: !Sub 'https://SignupAPI.execute−api.{SignupAPI}.execute-api.SignupAPI.execute−api.{AWS::Region}.amazonaws.com/prod/signup'

S3Bucket: Description: 'Your video/content storage bucket' Value: !Ref StreamingVideoBucket

UserPoolId: Description: 'Cognito User Pool ID - needed for frontend auth' Value: !Ref FixyachipsUserPool

UserPoolClientId: Description: 'Cognito Client ID - needed for frontend auth' Value: !Ref FixyachipsUserPoolClient

WAFArn: Description: 'WAF Web ACL ARN' Value: !GetAtt WebBodyguard.Arn ut and paste of the entire template:



Good. I have the **complete, untruncated YAML** now. Let me execute the full audit clean from scratch.

FIXYACHIPS GLOBAL v4.0 — COMPLETE STACK AUDIT

Principal AWS Serverless Architect + FinOps Security Review

VECTOR 1: COST OPTIMIZATION & FINOPS

- **FINOPS SAVINGS — S3: Zero Lifecycle Policy or Intelligent-Tiering Defined**

No lifecycle rules exist on `StreamingVideoBucket`. Every MP4, HLS segment, and DASH fragment ever uploaded sits in S3 Standard forever at full price regardless of viewership. For a VOD academy with a growing content library across 120 countries, this compounds monthly.

```
StreamingVideoBucket:
  Type: 'AWS::S3::Bucket'
  Properties:
    BucketName: !Sub 'fixyachips-platform-${AWS::AccountId}'
    PublicAccessBlockConfiguration:
      BlockPublicAcls: true
      BlockPublicPolicy: true
      IgnorePublicAcls: true
      RestrictPublicBuckets: true
    VersioningConfiguration:
      Status: Enabled
    IntelligentTieringConfigurations:
      - Id: VODIntelligentTiering
        Status: Enabled
        Tierings:
          - AccessTier: ARCHIVE_ACCESS
            Days: 90
          - AccessTier: DEEP_ARCHIVE_ACCESS
            Days: 180
```

```
    LifecycleConfiguration:
      Rules:
        - Id: TransitionVODToIntelligentTiering
          Status: Enabled
          Prefix: 'vod/'
          Transitions:
            - TransitionInDays: 30
              StorageClass: INTELLIGENT_TIERING
        - Id: TransitionHLSSegments
          Status: Enabled
          Prefix: 'hls/'
          Transitions:
            - TransitionInDays: 30
              StorageClass: INTELLIGENT_TIERING
        - Id: TransitionDASHSegments
          Status: Enabled
          Prefix: 'dash/'
          Transitions:
            - TransitionInDays: 30
              StorageClass: INTELLIGENT_TIERING
        - Id: AbortIncompleteUploads
          Status: Enabled
          AbortIncompleteMultipartUpload:
            DaysAfterInitiation: 7
    CorsConfiguration:
      CorsRules:
        - AllowedHeaders: ['*']
          AllowedMethods: [GET, HEAD]       # FIXED: removed POST/PUT - CDN is
read-only
          AllowedOrigins:
            - 'https://fixyachips.com'
            - 'https://www.fixyachips.com'
          ExposeHeaders:
            - 'Content-Length'
            - 'Content-Range'
            - 'ETag'
            - 'Accept-Ranges'
          MaxAge: 3600
```

- **FINOPS SAVINGS — S3 CORS Allows POST and PUT on a CDN Read-Only Bucket**

`AllowedMethods: [GET, POST, PUT]` on a bucket that CloudFront reads from is incorrect and a security/cost exposure. CloudFront never POSTs or PUTs to S3 on behalf of video viewers. Strip to `[GET, HEAD]` as shown above. HEAD is required for byte-range negotiation on video players.

- **FINOPS SAVINGS — Lambda MemorySize Not Set: Defaults to 128MB Cold Start Tax**

`SignupWorker` has no `MemorySize`. AWS defaults to 128MB. A Python 3.12 function cold-starting with boto3, DynamoDB, and two SES calls at 128MB runs 800ms–1.4s. At 256MB the same function runs in ~300ms. Lambda bills on GB-seconds — 256MB at 300ms costs the same or less than 128MB at 800ms, and your students get a faster signup confirmation.

```
SignupWorker:
```

```
  Type: 'AWS::Lambda::Function'
  Properties:
    FunctionName: !Sub 'FixyachipsSignupWorker-${AWS::StackName}'
    Handler: index.lambda_handler
    Runtime: python3.12
    Role: !GetAtt FixyachipsMasterRole.Arn
    Timeout: 15                          # REDUCED from 30: SES+DynamoDB never
needs 30s
    MemorySize: 256                      # ADDED: eliminates cold-start penalty
    ReservedConcurrentExecutions: 100    # ADDED: hard cap prevents runaway cost
during bot surge
    DeadLetterConfig:
      TargetArn: !GetAtt SignupWorkerDLQ.Arn   # ADDED: see DLQ resource below
    Environment:
      Variables:
        TABLE_NAME: !Ref StudentTable
        ADMIN_EMAIL: !Ref AdminEmail
```

- **FINOPS SAVINGS — CloudFront DefaultCacheBehavior Has No TTL Configuration**

`DefaultCacheBehavior` uses `ForwardedValues` (legacy method) with no `DefaultTTL`, `MinTTL`, or `MaxTTL` defined. CloudFront will honor whatever `Cache-Control` header S3 returns. If no header is set on your S3 objects, CloudFront applies a default of 24 hours for HTML — which is acceptable — but your VOD segments need 30-day TTLs to minimize Origin Shield cache misses billed at $0.0075–$0.012 per 10,000 requests. Add dedicated cache policies (required fix — see Vector 2 for the structural conflict).

- **FINOPS SAVINGS — Origin Shield Hardcoded to us-east-1 for a Primarily European Audience**

Your 8 languages are EN, ES, PT, FR, IT, DE, PL, SV. Six of eight are European. Origin Shield in `us-east-1` means every European CloudFront PoP that misses its regional cache fires a shield request across the Atlantic to Virginia. `eu-west-1` (Ireland) or `eu-central-1` (Frankfurt) cuts that latency and the per-request shield charge. Make it a parameter:

```
Parameters:
  AdminEmail:
    Type: String
    Default: 'info@fixyachips.com'
  OriginShieldRegion:
    Type: String
    Default: 'eu-west-1'
    AllowedValues:
      - 'us-east-1'
      - 'us-west-2'
      - 'eu-west-1'
      - 'eu-central-1'
      - 'ap-southeast-1'
      - 'ap-northeast-1'
    Description: 'Select the region closest to your highest-density viewer
concentration'
```

```
  CertificateArn:
    Type: String
    Default: ''
    Description: 'ACM Certificate ARN (must exist in us-east-1) for fixyachips.com'
  DomainName:
    Type: String
    Default: 'fixyachips.com'
  WWWDomainName:
    Type: String
    Default: 'www.fixyachips.com'
```

Then in the distribution:

```
OriginShield:
              Enabled: true
              OriginShieldRegion: !Ref OriginShieldRegion
```

- **FINOPS SAVINGS — WAF BotControl Has No Rate-Based Rule: Volumetric Bots Pass Free**

`WebBodyguard` catches known bots and common exploits but has no rate-based rule. A distributed botnet hitting `/signup` from 10,000 IPs at 1 req/sec each bypasses both managed rule groups entirely, hammers Lambda, fills DynamoDB with garbage leads, and exhausts your SES daily sending quota — getting your domain blacklisted. Each Lambda invocation and DynamoDB write costs real money at scale.

```
- Name: RateLimitSignupEndpoint
          Priority: 2
          Statement:
            RateBasedStatement:
              Limit: 1000
              AggregateKeyType: IP
              ScopeDownStatement:
                ByteMatchStatement:
                  SearchString: '/signup'
                  FieldToMatch:
                    UriPath: {}
                  TextTransformations:
                    - Priority: 0
                      Type: LOWERCASE
                  PositionalConstraint: CONTAINS
          Action:
            Block: {}
          VisibilityConfig:
            SampledRequestsEnabled: true
            CloudWatchMetricsEnabled: true
            MetricName: RateLimitSignup

        - Name: BlockSanctionedCountries
          Priority: 3
          Statement:
            GeoMatchStatement:
              CountryCodes: ['KP', 'IR', 'CU', 'SY']    # OFAC compliance
          Action:
            Block: {}
```

```
      VisibilityConfig:
        SampledRequestsEnabled: true
        CloudWatchMetricsEnabled: true
        MetricName: GeoBlock
```

VECTOR 2: SYNTAX, STRUCTURAL ERRORS & OMISSIONS

- **CRITICAL ERROR — WAF Rule Indentation: OverrideAction Is Outside ManagedRuleGroupStatement**

Both WAF rules have `OverrideAction` and `VisibilityConfig` at the wrong indentation level. They are inside `Statement` instead of at the `Rule` level. This will cause a CloudFormation deployment failure.

Current (BROKEN):

```
- Name: AWS-AWSManagedRulesCommonRuleSet
        Priority: 0
        Statement:
          ManagedRuleGroupStatement:
            VendorName: AWS
            Name: AWSManagedRulesCommonRuleSet
        OverrideAction: { None: {} }        # <- correct level
        VisibilityConfig:                   # <- correct level
```

Wait — re-reading the raw YAML carefully: `OverrideAction` IS at the rule level in your paste. However the **first rule** is missing `VisibilityConfig` at the rule level in the original PDF version. In the raw paste you provided, both rules are correctly structured. **This is confirmed clean in the raw YAML.** No fix needed on indentation for WAF.

- **CRITICAL ERROR — ForwardedValues and CachePolicyId Cannot Coexist**

`DefaultCacheBehavior` uses `ForwardedValues: QueryString: false`. This is the **legacy** CloudFront cache configuration method. You cannot add a `CachePolicyId` to this behavior without first removing `ForwardedValues`. This blocks you from ever implementing proper VOD TTL policies without a breaking stack update. Replace now:

```
# NEW RESOURCE: Add before WebsiteDistribution
WebCachePolicy:
  Type: 'AWS::CloudFront::CachePolicy'
  Properties:
    CachePolicyConfig:
      Name: !Sub 'FixyachipsWebCachePolicy-${AWS::StackName}'
      Comment: 'HTML/JS/CSS - short TTL for content updates'
      DefaultTTL: 86400
      MaxTTL: 604800
```

```
      MinTTL: 0
      ParametersInCacheKeyAndForwardedToOrigin:
        EnableAcceptEncodingGzip: true
        EnableAcceptEncodingBrotli: true
        HeadersConfig:
          HeaderBehavior: none
        CookiesConfig:
          CookieBehavior: none
        QueryStringsConfig:
          QueryStringBehavior: none

VODCachePolicy:
  Type: 'AWS::CloudFront::CachePolicy'
  Properties:
    CachePolicyConfig:
      Name: !Sub 'FixyachipsVODCachePolicy-${AWS::StackName}'
      Comment: 'HLS/DASH/MP4 segments - aggressive long-term caching'
      DefaultTTL: 2592000       # 30 days
      MaxTTL: 31536000          # 365 days
      MinTTL: 86400             # 1 day floor
      ParametersInCacheKeyAndForwardedToOrigin:
        EnableAcceptEncodingGzip: true
        EnableAcceptEncodingBrotli: true
        HeadersConfig:
          HeaderBehavior: none
        CookiesConfig:
          CookieBehavior: none
        QueryStringsConfig:
          QueryStringBehavior: none
```

- **CRITICAL ERROR — No Dead Letter Queue on SignupWorker Lambda**

`SignupWorker` is an internet-exposed async Lambda with zero DLQ. If DynamoDB throttles or SES rejects during a high-concurrency signup surge (launch day), the failure is silently swallowed. You lose real leads with no recovery path.

```
# NEW RESOURCE
SignupWorkerDLQ:
  Type: 'AWS::SQS::Queue'
  Properties:
    QueueName: !Sub 'FixyachipsSignupDLQ-${AWS::StackName}'
    MessageRetentionPeriod: 1209600    # 14 days maximum retention
    KmsMasterKeyId: alias/aws/sqs      # Encrypt dead letters at rest

# Add to FixyachipsMasterRole Policies Statement:
              - Effect: Allow
                Action:
                  - 'sqs:SendMessage'
                Resource: !GetAtt SignupWorkerDLQ.Arn
```

- **CRITICAL ERROR — CloudFront Distribution Missing Custom Domain and ACM Certificate**

`WebsiteDistribution` has no `Aliases` and no `ViewerCertificate`. The distribution serves from `*.cloudfront.net` — not `fixyachips.com`. Students in all 120 countries will see a raw CloudFront URL. Your brand does not exist at the CDN layer.

```
WebsiteDistribution:
    Type: 'AWS::CloudFront::Distribution'
    Properties:
      DistributionConfig:
        Enabled: true
        PriceClass: PriceClass_All
        DefaultRootObject: index.html
        Aliases:
          - !Ref DomainName
          - !Ref WWWDomainName
        ViewerCertificate:
          AcmCertificateArn: !Ref CertificateArn
          SslSupportMethod: sni-only
          MinimumProtocolVersion: TLSv1.2_2021    # Enforce modern TLS - required
for Smart TVs
        HttpVersion: http2and3                     # HTTP/3 for modern TV clients
        IPV6Enabled: true
        WebACLId: !GetAtt WebBodyguard.Arn
```

- **CRITICAL ERROR — CloudFront Function Runtime cloudfront-js-1.0 Is Deprecated**

`Runtime: 'cloudfront-js-1.0'` is the legacy runtime. `cloudfront-js-2.0` has been available since 2023, supports ES modules and async/await, and performs measurably better at 400+ global PoPs. Every viewer request in every country runs this function.

```
WebsiteRouterFunction:
    Type: 'AWS::CloudFront::Function'
    Properties:
      Name: !Sub 'FixRouter-${AWS::StackName}'
      AutoPublish: true
      FunctionConfig:
        Comment: 'Clean URLs + language path normalization for 8-language VOD
academy'
        Runtime: 'cloudfront-js-2.0'
      FunctionCode: |
        const SUPPORTED_LANGS = ['en','es','pt','fr','it','de','pl','sv'];
        async function handler(event) {
          const request = event.request;
          let uri = request.uri;
          const segments = uri.split('/').filter(s => s.length > 0);
          const first = segments[0] ? segments[0].toLowerCase() : '';
          if (!SUPPORTED_LANGS.includes(first)) {
            uri = '/en' + uri;
          }
          if (uri.endsWith('/')) {
            uri += 'index.html';
          } else if (!uri.includes('.')) {
            uri += '.html';
          }
          request.uri = uri;
          return request;
```

```
      }
```

- **CRITICAL ERROR — ApiOptionsMethod Uses AWS_PROXY for OPTIONS Preflight**

`ApiOptionsMethod` routes OPTIONS through Lambda via `AWS_PROXY`. Every browser preflight from every viewer in 120 countries cold-invokes a Lambda unnecessarily. OPTIONS should be handled by a `MOCK` integration returning static CORS headers — zero Lambda cost, zero latency penalty.

```
ApiOptionsMethod:
    Type: 'AWS::ApiGateway::Method'
    Properties:
      AuthorizationType: NONE
      HttpMethod: OPTIONS
      ResourceId: !Ref ApiResource
      RestApiId: !Ref SignupAPI
      Integration:
        Type: MOCK
        RequestTemplates:
          application/json: '{"statusCode": 200}'
        IntegrationResponses:
          - StatusCode: 200
            ResponseParameters:
              method.response.header.Access-Control-Allow-Headers: "'Content-
Type,X-Amz-Date,Authorization'"
              method.response.header.Access-Control-Allow-Methods: "'POST,OPTIONS'"
              method.response.header.Access-Control-Allow-Origin:
"'https://fixyachips.com'"
            ResponseTemplates:
              application/json: ''
      MethodResponses:
        - StatusCode: 200
          ResponseParameters:
            method.response.header.Access-Control-Allow-Headers: false
            method.response.header.Access-Control-Allow-Methods: false
            method.response.header.Access-Control-Allow-Origin: false
```

- **CRITICAL ERROR — API Gateway POST /signup Has No Throttling or Usage Plan**

`ApiMethod` has `AuthorizationType: NONE` and the `ApiDeployment` has no stage-level throttling. Any bot can hammer `/signup` at unlimited requests per second. No `UsagePlan`, no `ApiKey`, no `ThrottlingRateLimit` anywhere in the stack.

```
# REPLACE ApiDeployment with separate Stage resource:
ApiDeployment:
  Type: 'AWS::ApiGateway::Deployment'
  DependsOn:
    - ApiMethod
    - ApiOptionsMethod
  Properties:
    RestApiId: !Ref SignupAPI
    # StageName removed - managed by ApiStage below
```

```
ApiStage:
  Type: 'AWS::ApiGateway::Stage'
  Properties:
    RestApiId: !Ref SignupAPI
    DeploymentId: !Ref ApiDeployment
    StageName: prod
    MethodSettings:
      - ResourcePath: '/signup'
        HttpMethod: POST
        ThrottlingRateLimit: 10
        ThrottlingBurstLimit: 50
      - ResourcePath: '/*'
        HttpMethod: '*'
        ThrottlingRateLimit: 20
        ThrottlingBurstLimit: 100
    AccessLogSetting:
      DestinationArn: !GetAtt ApiAccessLogGroup.Arn

ApiAccessLogGroup:
  Type: 'AWS::Logs::LogGroup'
  Properties:
    LogGroupName: !Sub '/aws/apigateway/FixyachipsAPI-${AWS::StackName}'
    RetentionInDays: 30
```

Also update `LambdaPermission` SourceArn to reference the stage:

```
LambdaPermission:
    Type: 'AWS::Lambda::Permission'
    Properties:
      Action: 'lambda:InvokeFunction'
      FunctionName: !Ref SignupWorker
      Principal: apigateway.amazonaws.com
      SourceArn: !Sub 'arn:aws:execute-api:${AWS::Region}:${AWS::AccountId}:$
{SignupAPI}/prod/POST/signup'
```

- **ARCHITECTURAL ENHANCEMENT — DynamoDB Missing Point-in-Time Recovery, Email GSI, and Encryption**

`StudentTable` has `DeletionPolicy: Retain` (correct) but no PITR, no GSI on Email (making duplicate-check queries a full expensive Scan), and no encryption-at-rest specification.

```
StudentTable:
    Type: 'AWS::DynamoDB::Table'
    DeletionPolicy: Retain
    UpdateReplacePolicy: Retain
    Properties:
      TableName: !Sub 'FixyachipsStudents-${AWS::StackName}'
      AttributeDefinitions:
        - AttributeName: StudentID
          AttributeType: S
        - AttributeName: Email
          AttributeType: S
      KeySchema:
        - AttributeName: StudentID
          KeyType: HASH
      GlobalSecondaryIndexes:
```

```
        - IndexName: EmailIndex
          KeySchema:
            - AttributeName: Email
              KeyType: HASH
          Projection:
            ProjectionType: ALL
      BillingMode: PAY_PER_REQUEST
      PointInTimeRecoverySpecification:
        PointInTimeRecoveryEnabled: true
      SSESpecification:
        SSEEnabled: true
        SSEType: KMS
```

Also update the IAM role to allow GSI queries:

```
- Effect: Allow
                Action:
                  - 'dynamodb:GetItem'
                  - 'dynamodb:PutItem'
                  - 'dynamodb:UpdateItem'
                  - 'dynamodb:DeleteItem'
                  - 'dynamodb:Query'
                  - 'dynamodb:Scan'
                Resource:
                  - !GetAtt StudentTable.Arn
                  - !Sub '${StudentTable.Arn}/index/EmailIndex'
```

- **ARCHITECTURAL ENHANCEMENT — CloudFront Has No Access Logging**

Zero `Logging` configuration on `WebsiteDistribution`. Without access logs you have no visibility into cache hit ratios per language path, geographic request patterns, or VOD segment playback failures across 120 countries.

```
# NEW RESOURCE
CloudFrontLogsBucket:
  Type: 'AWS::S3::Bucket'
  Properties:
    BucketName: !Sub 'fixyachips-cf-logs-${AWS::AccountId}'
    PublicAccessBlockConfiguration:
      BlockPublicAcls: true
      BlockPublicPolicy: true
      IgnorePublicAcls: true
      RestrictPublicBuckets: true
    OwnershipControls:
      Rules:
        - ObjectOwnership: BucketOwnerPreferred
    LifecycleConfiguration:
      Rules:
        - Id: ExpireOldLogs
          Status: Enabled
          ExpirationInDays: 90

# Add inside DistributionConfig:
        Logging:
          Bucket: !GetAtt CloudFrontLogsBucket.DomainName
          Prefix: 'cloudfront-logs/'
```

```
          IncludeCookies: false
```

- **ARCHITECTURAL ENHANCEMENT — Missing CloudFront CustomErrorResponses**

No `CustomErrorResponses` defined. When a video segment URL is wrong or a CloudFront cache miss hits a private S3 object, CloudFront returns a raw XML `AccessDenied` response. Smart TVs, Roku sticks, and Fire TV clients cannot parse XML error pages — they freeze or display a black screen with no error to the student.

```
CustomErrorResponses:
          - ErrorCode: 403
            ResponseCode: 200
            ResponsePagePath: '/index.html'
            ErrorCachingMinTTL: 10
          - ErrorCode: 404
            ResponseCode: 200
            ResponsePagePath: '/index.html'
            ErrorCachingMinTTL: 10
          - ErrorCode: 500
            ResponseCode: 503
            ResponsePagePath: '/index.html'
            ErrorCachingMinTTL: 5
          - ErrorCode: 503
            ResponseCode: 503
            ResponsePagePath: '/index.html'
            ErrorCachingMinTTL: 5
```

VECTOR 3: WEB-TO-TV STREAM CONVERSION & VOD DELIVERY

- **CRITICAL ERROR — HTTP Range Requests Not Configured: Video Scrubbing Broken on All TV Clients**

There is no `OriginRequestPolicy`, no `ResponseHeadersPolicy`, and no `AllowedMethods` that includes `OPTIONS` in the `DefaultCacheBehavior`. The `Range` header is not forwarded to S3. This means:

- A student on Samsung TV scrubbing to minute 45 of a 60-minute lesson issues a `Range: bytes=...` request
- CloudFront strips the Range header (default behavior without explicit forwarding)
- S3 returns the entire file from byte 0
- The TV client times out or buffers infinitely

This is a complete VOD delivery failure for every non-browser client.

```
# NEW RESOURCE
VODOriginRequestPolicy:
```

```
  Type: 'AWS::CloudFront::OriginRequestPolicy'
  Properties:
    OriginRequestPolicyConfig:
      Name: !Sub 'FixyachipsVODOriginRequest-${AWS::StackName}'
      Comment: 'Forward Range header for VOD byte-range serving to all TV clients'
      HeadersConfig:
        HeaderBehavior: whitelist
        Headers:
          - 'Origin'
          - 'Access-Control-Request-Headers'
          - 'Access-Control-Request-Method'
          - 'Range'
      CookiesConfig:
        CookieBehavior: none
      QueryStringsConfig:
        QueryStringBehavior: none

# NEW RESOURCE
VODResponseHeadersPolicy:
  Type: 'AWS::CloudFront::ResponseHeadersPolicy'
  Properties:
    ResponseHeadersPolicyConfig:
      Name: !Sub 'FixyachipsVODResponseHeaders-${AWS::StackName}'
      Comment: 'CORS + security headers + byte-range signals for VOD delivery'
      CorsConfig:
        AccessControlAllowCredentials: false
        AccessControlAllowHeaders:
          Items: ['*']
        AccessControlAllowMethods:
          Items: [GET, HEAD, OPTIONS]
        AccessControlAllowOrigins:
          Items:
            - 'https://fixyachips.com'
            - 'https://www.fixyachips.com'
        AccessControlExposeHeaders:
          Items:
            - 'Content-Length'
            - 'Content-Range'      # CRITICAL: player tracks playback position
            - 'ETag'
            - 'Accept-Ranges'      # CRITICAL: signals byte-range support to TV
clients
        AccessControlMaxAgeSec: 3600
        OriginOverride: false
      SecurityHeadersConfig:
        ContentTypeOptions:
          Override: true
        FrameOptions:
          FrameOption: DENY
          Override: true
        ReferrerPolicy:
          ReferrerPolicy: strict-origin-when-cross-origin
          Override: true
        StrictTransportSecurity:
          AccessControlMaxAgeSec: 31536000
          IncludeSubdomains: true
          Preload: true
          Override: true
        XSSProtection:
          ModeBlock: true
```

```
          Override: true
          Protection: true
```

- **CRITICAL ERROR — No CacheBehaviors for VOD Paths: Video Segments Served with HTML TTLs**

There is only a `DefaultCacheBehavior`. Zero `CacheBehaviors` are defined. Your HLS `.m3u8` manifests, `.ts` segments, DASH `.mpd` manifests, `.m4s` fragments, and MP4 files all get served with the same policy as your homepage HTML. HLS manifests should cache for 5–10 seconds (live-adjacent playlists) or hours (static VOD). Segments should cache for 30 days. HTML should cache for 1 day. One policy for all of these is architecturally broken.

```
# Add inside DistributionConfig, after DefaultCacheBehavior:
        CacheBehaviors:
          - PathPattern: '/hls/*'
            TargetOriginId: S3Origin
            ViewerProtocolPolicy: redirect-to-https
            Compress: false                    # Never compress pre-encoded video
            CachePolicyId: !Ref VODCachePolicy
            OriginRequestPolicyId: !Ref VODOriginRequestPolicy
            ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
            AllowedMethods: [GET, HEAD, OPTIONS]
            CachedMethods: [GET, HEAD]
          - PathPattern: '/dash/*'
            TargetOriginId: S3Origin
            ViewerProtocolPolicy: redirect-to-https
            Compress: false
            CachePolicyId: !Ref VODCachePolicy
            OriginRequestPolicyId: !Ref VODOriginRequestPolicy
            ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
            AllowedMethods: [GET, HEAD, OPTIONS]
            CachedMethods: [GET, HEAD]
          - PathPattern: '/vod/*'
            TargetOriginId: S3Origin
            ViewerProtocolPolicy: redirect-to-https
            Compress: false
            CachePolicyId: !Ref VODCachePolicy
            OriginRequestPolicyId: !Ref VODOriginRequestPolicy
            ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
            AllowedMethods: [GET, HEAD, OPTIONS]
            CachedMethods: [GET, HEAD]
```

- **ARCHITECTURAL ENHANCEMENT — No MediaConvert Pipeline: Raw MP4 Is Not Production VOD**

The template stores MP4 files in S3 but has zero AWS Elemental MediaConvert integration. Raw unencoded MP4 cannot serve adaptive bitrate streaming. A student in rural Poland on a 3G connection gets the same 1080p stream as a student in Stockholm on fiber — the Polish student buffers constantly and abandons the course. You need HLS/DASH multi-bitrate renditions (240p/480p/720p/1080p) generated by MediaConvert on upload.

```
# NEW RESOURCE
MediaConvertRole:
  Type: 'AWS::IAM::Role'
  Properties:
    RoleName: !Sub 'FixyachipsMediaConvertRole-${AWS::StackName}'
    AssumeRolePolicyDocument:
      Version: '2012-10-17'
      Statement:
        - Effect: Allow
          Principal:
            Service: mediaconvert.amazonaws.com
          Action: 'sts:AssumeRole'
    Policies:
      - PolicyName: MediaConvertS3Access
        PolicyDocument:
          Version: '2012-10-17'
          Statement:
            - Effect: Allow
              Action:
                - 's3:GetObject'
                - 's3:PutObject'
              Resource:
                - !Sub 'arn:aws:s3:::${StreamingVideoBucket}/raw-uploads/*'
                - !Sub 'arn:aws:s3:::${StreamingVideoBucket}/vod/*'
                - !Sub 'arn:aws:s3:::${StreamingVideoBucket}/hls/*'
                - !Sub 'arn:aws:s3:::${StreamingVideoBucket}/dash/*'

# NEW RESOURCE
VideoProcessingFunction:
  Type: 'AWS::Lambda::Function'
  Properties:
    FunctionName: !Sub 'FixyachipsVideoProcessor-${AWS::StackName}'
    Handler: index.lambda_handler
    Runtime: python3.12
    Role: !GetAtt FixyachipsMasterRole.Arn
    MemorySize: 512
    Timeout: 60
    Environment:
      Variables:
        MEDIACONVERT_ROLE_ARN: !GetAtt MediaConvertRole.Arn
        OUTPUT_BUCKET: !Ref StreamingVideoBucket
        MEDIACONVERT_ENDPOINT: ''    # Populate via SSM Parameter or custom
resource
    Code:
      ZipFile: |
        import json, boto3, os, re
        def lambda_handler(event, context):
            endpoint = os.environ['MEDIACONVERT_ENDPOINT']
            mc = boto3.client('mediaconvert', endpoint_url=endpoint)
            for record in event['Records']:
                key = record['s3']['object']['key']
                if not key.startswith('raw-uploads/'):
                    continue
                bucket = record['s3']['bucket']['name']
                base = re.sub(r'^raw-uploads/(.+)\.[^.]+$', r'\1', key)
                mc.create_job(
                    Role=os.environ['MEDIACONVERT_ROLE_ARN'],
                    Settings={
                        'Inputs': [{'FileInput': f's3://{bucket}/{key}'}],
```

```
                        'OutputGroups': [{
                            'Name': 'HLS',
                            'OutputGroupSettings': {
                                'Type': 'HLS_GROUP_SETTINGS',
                                'HlsGroupSettings': {
                                    'Destination':
f"s3://{os.environ['OUTPUT_BUCKET']}/hls/{base}/",
                                    'SegmentLength': 6
                                }
                            },
                            'Outputs': [
                                {'NameModifier': '_1080p', 'VideoDescription':
{'Width': 1920, 'Height': 1080}},
                                {'NameModifier': '_720p',  'VideoDescription':
{'Width': 1280, 'Height': 720}},
                                {'NameModifier': '_480p',  'VideoDescription':
{'Width': 854,  'Height': 480}},
                                {'NameModifier': '_240p',  'VideoDescription':
{'Width': 426,  'Height': 240}}
                            ]
                        }]
                    }
                )
                print(f'MediaConvert job submitted: {key}')

# NEW RESOURCE - EventBridge trigger for S3 uploads
VideoUploadEventRule:
  Type: 'AWS::Events::Rule'
  Properties:
    Name: !Sub 'FixyachipsVideoUpload-${AWS::StackName}'
    EventPattern:
      source: ['aws.s3']
      detail-type: ['Object Created']
      detail:
        bucket:
          name: [!Ref StreamingVideoBucket]
        object:
          key:
            - prefix: 'raw-uploads/'
    Targets:
      - Id: VideoProcessingLambda
        Arn: !GetAtt VideoProcessingFunction.Arn

VideoProcessorEventPermission:
  Type: 'AWS::Lambda::Permission'
  Properties:
    Action: 'lambda:InvokeFunction'
    FunctionName: !Ref VideoProcessingFunction
    Principal: events.amazonaws.com
    SourceArn: !GetAtt VideoUploadEventRule.Arn
```

VECTOR 4: MULTI-LINGUAL ARCHITECTURE

- **CRITICAL ERROR — Zero Language Routing Architecture Exists in the Entire Template**

The `WebsiteRouterFunction` only appends `.html` to clean URLs. It has no language awareness. There are no `CacheBehaviors` for `/es/*`, `/de/*`, `/fr/*`, `/it/*`, `/pt/*`, `/pl/*`, `/sv/*`. There is no mechanism to route 8 languages to 8 separate S3 path prefixes. The entire multi-lingual claim in the stack description has zero infrastructure backing it.

The corrected `WebsiteRouterFunction` in Vector 2 above handles this. The corrected `CacheBehaviors` below complete it:

```
# Add to CacheBehaviors list (in addition to /hls/*, /dash/*, /vod/*):
          - PathPattern: '/es/*'
            TargetOriginId: S3Origin
            ViewerProtocolPolicy: redirect-to-https
            Compress: true
            CachePolicyId: !Ref WebCachePolicy
            ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
            AllowedMethods: [GET, HEAD]
            CachedMethods: [GET, HEAD]
            FunctionAssociations:
              - EventType: viewer-request
                FunctionARN: !GetAtt
WebsiteRouterFunction.FunctionMetadata.FunctionARN
          - PathPattern: '/pt/*'
            TargetOriginId: S3Origin
            ViewerProtocolPolicy: redirect-to-https
            Compress: true
            CachePolicyId: !Ref WebCachePolicy
            ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
            AllowedMethods: [GET, HEAD]
            CachedMethods: [GET, HEAD]
            FunctionAssociations:
              - EventType: viewer-request
                FunctionARN: !GetAtt
WebsiteRouterFunction.FunctionMetadata.FunctionARN
          - PathPattern: '/fr/*'
            TargetOriginId: S3Origin
            ViewerProtocolPolicy: redirect-to-https
            Compress: true
            CachePolicyId: !Ref WebCachePolicy
            ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
            AllowedMethods: [GET, HEAD]
            CachedMethods: [GET, HEAD]
            FunctionAssociations:
              - EventType: viewer-request
                FunctionARN: !GetAtt
WebsiteRouterFunction.FunctionMetadata.FunctionARN
          - PathPattern: '/it/*'
            TargetOriginId: S3Origin
            ViewerProtocolPolicy: redirect-to-https
            Compress: true
            CachePolicyId: !Ref WebCachePolicy
            ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
            AllowedMethods: [GET, HEAD]
            CachedMethods: [GET, HEAD]
            FunctionAssociations:
```

```
              - EventType: viewer-request
                FunctionARN: !GetAtt
WebsiteRouterFunction.FunctionMetadata.FunctionARN
          - PathPattern: '/de/*'
            TargetOriginId: S3Origin
            ViewerProtocolPolicy: redirect-to-https
            Compress: true
            CachePolicyId: !Ref WebCachePolicy
            ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
            AllowedMethods: [GET, HEAD]
            CachedMethods: [GET, HEAD]
            FunctionAssociations:
              - EventType: viewer-request
                FunctionARN: !GetAtt
WebsiteRouterFunction.FunctionMetadata.FunctionARN
          - PathPattern: '/pl/*'
            TargetOriginId: S3Origin
            ViewerProtocolPolicy: redirect-to-https
            Compress: true
            CachePolicyId: !Ref WebCachePolicy
            ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
            AllowedMethods: [GET, HEAD]
            CachedMethods: [GET, HEAD]
            FunctionAssociations:
              - EventType: viewer-request
                FunctionARN: !GetAtt
WebsiteRouterFunction.FunctionMetadata.FunctionARN
          - PathPattern: '/sv/*'
            TargetOriginId: S3Origin
            ViewerProtocolPolicy: redirect-to-https
            Compress: true
            CachePolicyId: !Ref WebCachePolicy
            ResponseHeadersPolicyId: !Ref VODResponseHeadersPolicy
            AllowedMethods: [GET, HEAD]
            CachedMethods: [GET, HEAD]
            FunctionAssociations:
              - EventType: viewer-request
                FunctionARN: !GetAtt
WebsiteRouterFunction.FunctionMetadata.FunctionARN
```

- **CRITICAL ERROR — Do NOT Forward Accept-Language Header in Cache Key**

This is the most common cache-destruction mistake in multi-lingual CloudFront deployments. If `Accept-Language` is forwarded in the cache key, Chrome on Windows sends `de-DE,de;q=0.9,en;q=0.8` while Firefox sends `de,en-US;q=0.7,en;q=0.3`. These are two different cache keys for the identical `/de/index.html` page. Your global cache hit ratio collapses to near zero and every request pays Origin Shield charges.

The correct architecture is path-based routing exclusively. Content stored at `/de/`, `/es/`, `/fr/` in S3. CloudFront caches by path. Zero header forwarding. The CloudFront Function transparently prepends the language prefix. This is already implemented in the corrected function above.

- **ARCHITECTURAL ENHANCEMENT — Recommended S3 Key Namespace for 8 Languages**

Enforce this structure via your CI/CD upload pipeline and MediaConvert output paths:

```
# Static web content
/en/index.html
/es/index.html
/de/index.html
/fr/index.html
/it/index.html
/pt/index.html
/pl/index.html
/sv/index.html

# HLS per language per course
/hls/en/{course-id}/master.m3u8
/hls/es/{course-id}/master.m3u8
/hls/de/{course-id}/master.m3u8

# DASH per language per course
/dash/en/{course-id}/manifest.mpd
/dash/es/{course-id}/manifest.mpd

# Raw upload input (MediaConvert source)
/raw-uploads/{course-id}-{lang}.mp4
```

VECTOR 5: SECURITY BLINDSPOTS

- **CRITICAL ERROR — bedrock:* with Resource: * on an Internet-Exposed Lambda Role**

This is the single most dangerous line in the template. `FixyachipsMasterRole` grants `bedrock:*` on `Resource: '*'` to `SignupWorker` — a Lambda that accepts unauthenticated POST requests from the public internet via API Gateway. If an attacker finds a code injection vector, they can invoke Claude 3 Opus at $15/million tokens in an infinite loop. Your AWS bill could hit $10,000 before a CloudWatch alarm fires. Scope all AI/ML permissions to minimum required actions only.

```
# REPLACE the entire AI/ML block with this:
              - Effect: Allow
                Action:
                  - 'ses:SendEmail'
                  - 'ses:SendRawEmail'
                Resource:
                  - !Sub
'arn:aws:ses:${AWS::Region}:${AWS::AccountId}:identity/fixyachips.com'
                  - !Sub 'arn:aws:ses:${AWS::Region}:${AWS::AccountId}:identity/$
{AdminEmail}'
              # Bedrock: specific models only - NEVER bedrock:*
              - Effect: Allow
                Action:
```

```
                - 'bedrock:InvokeModel'
              Resource:
                - !Sub
'arn:aws:bedrock:${AWS::Region}::foundation-model/amazon.titan-text-express-v1'
            # Translate: no resource-level permissions available (AWS limitation)
            - Effect: Allow
              Action:
                - 'translate:TranslateText'
              Resource: '*'
            # Polly: no resource-level permissions available (AWS limitation)
            - Effect: Allow
              Action:
                - 'polly:SynthesizeSpeech'
              Resource: '*'
            # Transcribe: job-level only
            - Effect: Allow
              Action:
                - 'transcribe:StartTranscriptionJob'
                - 'transcribe:GetTranscriptionJob'
              Resource: '*'
```

- **CRITICAL ERROR — Cognito Client Has ALLOW_USER_PASSWORD_AUTH Enabled**

`ALLOW_USER_PASSWORD_AUTH` transmits the raw password to the Cognito endpoint. For a public SPA this is the weakest auth flow available and is explicitly flagged in AWS security documentation. Use `ALLOW_USER_SRP_AUTH` exclusively — the password never leaves the client in plaintext under SRP.

```
FixyachipsUserPoolClient:
    Type: 'AWS::Cognito::UserPoolClient'
    Properties:
      ClientName: Fixyachips_Web_Portal
      UserPoolId: !Ref FixyachipsUserPool
      GenerateSecret: false
      ExplicitAuthFlows:
        - ALLOW_USER_SRP_AUTH          # KEEP: secure zero-knowledge password proof
        - ALLOW_REFRESH_TOKEN_AUTH     # KEEP: token refresh
        # REMOVED: ALLOW_USER_PASSWORD_AUTH
      PreventUserExistenceErrors: ENABLED    # ADDED: don't leak whether email is
registered
      RefreshTokenValidity: 30
      AccessTokenValidity: 1
      IdTokenValidity: 1
      TokenValidityUnits:
        RefreshToken: days
        AccessToken: hours
        IdToken: hours
      EnableTokenRevocation: true
```

- **CRITICAL ERROR — Cognito UserPool Has No MFA and No Advanced Security**

`FixyachipsUserPool` has a strong password policy but zero MFA configuration and no `UserPoolAddOns`. For a paid VOD academy, credential stuffing attacks (automated login attempts using breached password databases) are the #1 account takeover vector. Without Advanced Security Mode, Cognito does not block known-compromised credentials.

```
FixyachipsUserPool:
    Type: 'AWS::Cognito::UserPool'
    Properties:
      UserPoolName: !Sub 'Fixyachips_Global_Users-${AWS::StackName}'
      UsernameAttributes: [email]
      AutoVerifiedAttributes: [email]
      MfaConfiguration: OPTIONAL
      EnabledMfas:
        - TOTP
      UserPoolAddOns:
        AdvancedSecurityMode: ENFORCED    # Blocks compromised credentials
automatically
      AccountRecoverySetting:
        RecoveryMechanisms:
          - Name: verified_email
            Priority: 1
      Policies:
        PasswordPolicy:
          MinimumLength: 8
          RequireLowercase: true
          RequireNumbers: true
          RequireSymbols: true
          RequireUppercase: true
          TemporaryPasswordValidityDays: 7
      Schema:
        - Name: email
          AttributeDataType: String
          Mutable: true
          Required: true
        - Name: veteran_status
          AttributeDataType: String
          Mutable: true
        - Name: industry_segment
          AttributeDataType: String
          Mutable: true
```

- **CRITICAL ERROR — Lambda SignupWorker Has No Duplicate Email Check**

`table.put_item()` is called unconditionally. A student submitting the form twice creates two `StudentID` records and triggers two SES welcome emails. Two emails from the same sender address within seconds of each other is a spam signal that damages your SES domain reputation. Use the `EmailIndex` GSI added in Vector 2:

```
# Add this block inside lambda_handler BEFORE table.put_item():

                # Check for duplicate email using GSI
                existing = table.query(
                    IndexName='EmailIndex',
                    KeyConditionExpression='Email = :email',
```

```
                    ExpressionAttributeValues={':email': email},
                    Limit=1
                )
                if existing.get('Count', 0) > 0:
                    return {
                        'statusCode': 200,
                        'headers': {
                            'Access-Control-Allow-Origin':
'https://fixyachips.com',
                            'Content-Type': 'application/json'
                        },
                        'body': json.dumps({
                            'status': 'already_registered',
                            'message': 'You are already on the list. We will
notify you!'
                        })
                    }
```

- **CRITICAL ERROR — No CloudWatch Alarms: Zero Operational Visibility**

There are zero alarms in the entire stack. No WAF block rate alarm, no Lambda error alarm, no DynamoDB throttle alarm. For a global platform serving 120 countries, you will discover attacks and outages from student complaints, not from your monitoring system.

```
# NEW RESOURCES
SecurityAlertTopic:
  Type: 'AWS::SNS::Topic'
  Properties:
    TopicName: !Sub 'FixyachipsSecurityAlerts-${AWS::StackName}'
    Subscription:
      - Protocol: email
        Endpoint: !Ref AdminEmail

WAFBlocksAlarm:
  Type: 'AWS::CloudWatch::Alarm'
  Properties:
    AlarmName: !Sub 'FixyachipsWAFHighBlocks-${AWS::StackName}'
    AlarmDescription: 'WAF blocking abnormal volume - potential attack in progress'
    MetricName: BlockedRequests
    Namespace: AWS/WAFV2
    Dimensions:
      - Name: WebACL
        Value: !Sub 'FixyachipsShield-${AWS::StackName}'
      - Name: Region
        Value: 'us-east-1'
      - Name: Rule
        Value: ALL
    Statistic: Sum
    Period: 300
    EvaluationPeriods: 1
    Threshold: 1000
    ComparisonOperator: GreaterThanThreshold
    AlarmActions: [!Ref SecurityAlertTopic]
    TreatMissingData: notBreaching

LambdaErrorAlarm:
```

```
  Type: 'AWS::CloudWatch::Alarm'
  Properties:
    AlarmName: !Sub 'FixyachipsSignupErrors-${AWS::StackName}'
    AlarmDescription: 'SignupWorker Lambda error rate elevated'
    MetricName: Errors
    Namespace: AWS/Lambda
    Dimensions:
      - Name: FunctionName
        Value: !Ref SignupWorker
    Statistic: Sum
    Period: 300
    EvaluationPeriods: 1
    Threshold: 10
    ComparisonOperator: GreaterThanThreshold
    AlarmActions: [!Ref SecurityAlertTopic]
    TreatMissingData: notBreaching

SignupVolumeAlarm:
  Type: 'AWS::CloudWatch::Alarm'
  Properties:
    AlarmName: !Sub 'FixyachipsSignupAnomaly-${AWS::StackName}'
    AlarmDescription: 'Unusual signup volume - possible bot surge or viral event'
    MetricName: Invocations
    Namespace: AWS/Lambda
    Dimensions:
      - Name: FunctionName
        Value: !Ref SignupWorker
    Statistic: Sum
    Period: 300
    EvaluationPeriods: 1
    Threshold: 500
    ComparisonOperator: GreaterThanThreshold
    AlarmActions: [!Ref SecurityAlertTopic]
    TreatMissingData: notBreaching
```

- **ARCHITECTURAL ENHANCEMENT — No Geo-Restriction Structure for Licensing Readiness**

The distribution has no `Restrictions` block. If fixyachips.com acquires regional content licensing restrictions (standard in VOD/education at scale), there is no enforcement mechanism. Add the scaffolding now so it activates with a single parameter change rather than a structural stack update:

```
# Add inside DistributionConfig:
        Restrictions:
          GeoRestriction:
            RestrictionType: none
            # To activate blacklist: change to 'blacklist' and add:
            # Locations: ['KP', 'IR', 'CU', 'SY']
```

- **ARCHITECTURAL ENHANCEMENT — Missing Outputs for Language URLs, DLQ ARN, and Distribution ID**

Current outputs expose the CloudFront domain but not the distribution ID (needed for cache invalidations), the DLQ ARN (needed for operations), or language-specific URLs (needed for SEO hreflang tags and frontend routing).

```
DistributionId:
    Description: 'CloudFront Distribution ID - use for cache invalidations'
    Value: !Ref WebsiteDistribution

  DLQUrl:
    Description: 'Signup Worker Dead Letter Queue - monitor for failed signups'
    Value: !Ref SignupWorkerDLQ

  LanguageBaseEN:
    Value: !Sub 'https://${DomainName}/en/'
  LanguageBaseES:
    Value: !Sub 'https://${DomainName}/es/'
  LanguageBaseDE:
    Value: !Sub 'https://${DomainName}/de/'
  LanguageBaseFR:
    Value: !Sub 'https://${DomainName}/fr/'
  LanguageBasePT:
    Value: !Sub 'https://${DomainName}/pt/'
  LanguageBaseIT:
    Value: !Sub 'https://${DomainName}/it/'
  LanguageBasePL:
    Value: !Sub 'https://${DomainName}/pl/'
  LanguageBaseSV:
    Value: !Sub 'https://${DomainName}/sv/'
```

MASTER SEVERITY SUMMARY

#	Finding	Severity	Vector
1	`bedrock:*` wildcard on internet-exposed public Lambda	**CRITICAL**	5
2	No ACM certificate / custom domain on CloudFront distribution	**CRITICAL**	2
3	`ForwardedValues` blocks cache policy adoption permanently	**CRITICAL**	2
4	Range header not forwarded — video scrubbing broken on all TV clients	**CRITICAL**	3
5	OPTIONS method using Lambda proxy — CORS preflight hits Lambda unnecessarily	**CRITICAL**	2
6	No DLQ on SignupWorker — leads silently lost during surge	**CRITICAL**	2
7	No API Gateway throttling — unlimited bot requests to /signup	**CRITICAL**	5
8	`ALLOW_USER_PASSWORD_AUTH` on Cognito — plaintext password risk	**CRITICAL**	5
9	Zero language routing architecture for 8 languages	**CRITICAL**	4
10	No VOD CacheBehaviors — video served with HTML TTLs	**CRITICAL**	3
11	CloudFront Function on deprecated JS 1.0 runtime	**CRITICAL**	2
12	No WAF rate-based rule — volumetric attacks bypass all managed rules	**CRITICAL**	5

#	Finding	Severity	Vector
13	No duplicate email check — double signups + SES reputation damage	**CRITICAL**	5
14	Cognito missing MFA and AdvancedSecurityMode	**CRITICAL**	5
15	S3 lifecycle / Intelligent-Tiering absent on VOD assets	**FINOPS**	1
16	Lambda memory unset — 128MB cold start billing tax	**FINOPS**	1
17	Lambda timeout 30s — 2x correct value, billing waste	**FINOPS**	1
18	Origin Shield hardcoded us-east-1 for European-majority audience	**FINOPS**	1
19	No TTL config — CloudFront falls back to S3 origin headers	**FINOPS**	1
20	S3 CORS allows POST/PUT on a read-only CDN bucket	**FINOPS**	1
21	No MediaConvert pipeline — raw MP4 not ABR encoded for global delivery	**ARCHITECTURAL**	3
22	No CloudFront access logging — zero visibility into 120-country traffic	**ARCHITECTURAL**	2
23	DynamoDB missing PITR, Email GSI, and encryption-at-rest	**ARCHITECTURAL**	2
24	Zero CloudWatch alarms — attacks discovered from student complaints	**ARCHITECTURAL**	5
25	No S3 bucket versioning — accidental video overwrites unrecoverable	**ARCHITECTURAL**	5
26	No geo-restriction scaffolding for licensing readiness	**ARCHITECTURAL**	5
27	HTTP/3 not enabled — modern Smart TV clients cannot use QUIC	**ARCHITECTURAL**	3
28	No CustomErrorResponses — TV clients receive raw XML S3 errors	**ARCHITECTURAL**	3
29	Missing distribution ID, DLQ ARN, and language URL outputs	**ARCHITECTURAL**	2